

FRAME MODELER 2D

**An Open-Source Platform for the 2-Dimensional Modeling and Seismic
Analysis of Buildings**

Version 5.2606

Manual

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Citation

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Download and Support Links

GitHub repository: <https://github.com/amaelkady/FM-2D>

Registration for software updates: <https://forms.gle/qLNwgGFS5vbCxmHB7>

Video tutorials on YouTube: https://www.youtube.com/playlist?list=PLz_XdUL-6Y_m10fBEMmzPKqI00wdzbDUN

1. Installation

The step-by-step installation procedure of the software, in Windows, is discussed in this section. Note that FM-2D executable is only made available for Windows machines, however, the software can run directly on a Mac machine using the MATLAB source code. Figure 1.1 shows all the files/folder that are downloaded by the user through the FM-2D's GitHub repository. These are as follows:

- 2) *FM2D_Installer.exe* is the main installation file.
- 3) *FM2D Manual.pdf* is FM-2D's manual and installation guide.
- 4) *Supporting Documents* this folder includes all the pre-formatted and template files that can be modified by the user and imported into the different modules in FM-2D.

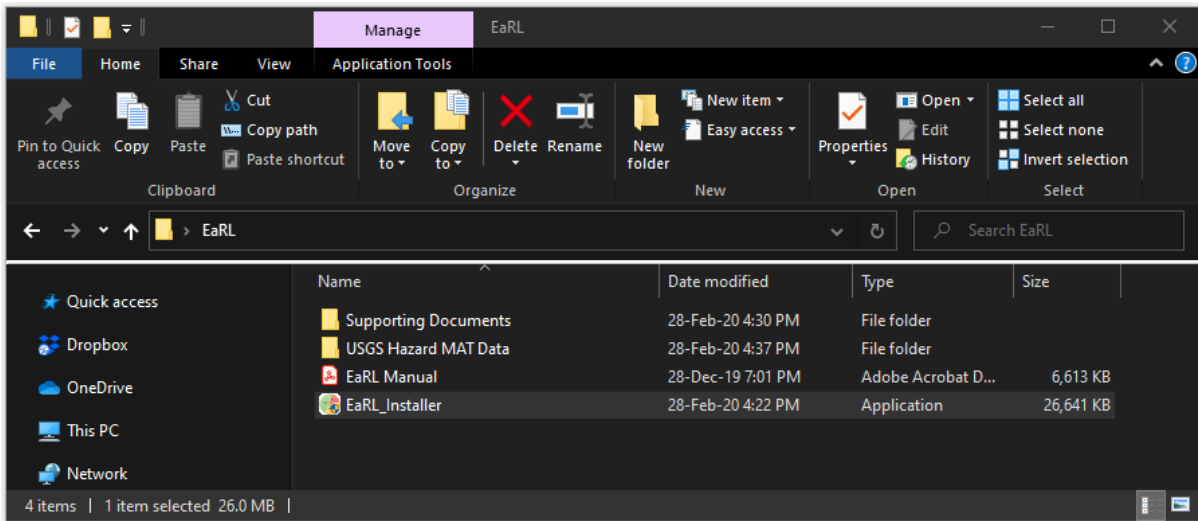


Figure 1.1. Downloaded software files and folders

The software can be installed by running the *FM2D_Installer.exe* executable file. As demonstrated in Figure 1.2, the installation process is similar to any software. The user is prompted to different windows to approve the installation process and set the installation directory. By default, the installation directory is “C:\Program Files\University of Southampton\FM_2D”. Note that if MATLAB runtime was not already installed on the local PC, the user will be prompted to download the latest runtime environment (about 700Mb). An internet connection is required in this case.

Once installed, the executable file *FM_2D.exe*, as well as other files, can be found in the default installation directory under “.....\University of Southampton\FM_2D\application\FM_2D.exe” as shown in Figure 1.3. Since FM-2D runs analysis using *OpenSees*, users shall independently install Tcl/Tk and *OpenSees* on their local PC. *OpenSees* installation guide can be found at <https://opensees.berkeley.edu/OpenSees/user/download.php>. After installation, the user must make sure the executable *OpenSees.exe*, which was already generated during FM-2D installation procedure, is working properly from the installation directory. If not, then please re-check your *OpenSees* installation procedure and note that in Windows machines the *OpenSees* “lib” folder would typically need to be located within the “...\University of Southampton\FM_2D” directory. Finally, it is important to note that FM-2D should be opened by running the *FM_2D.exe* file from the installation directory using administrator privileges. Otherwise, internal processes that require reading and writing files, in the installation directory, may not work properly.

The users are highly recommended to register their information online using the following form link: <https://forms.gle/CkfSBmjhqATJZQJj6>; so that they are able to receive e-mails notification regarding future releases and/or bug-fixes.

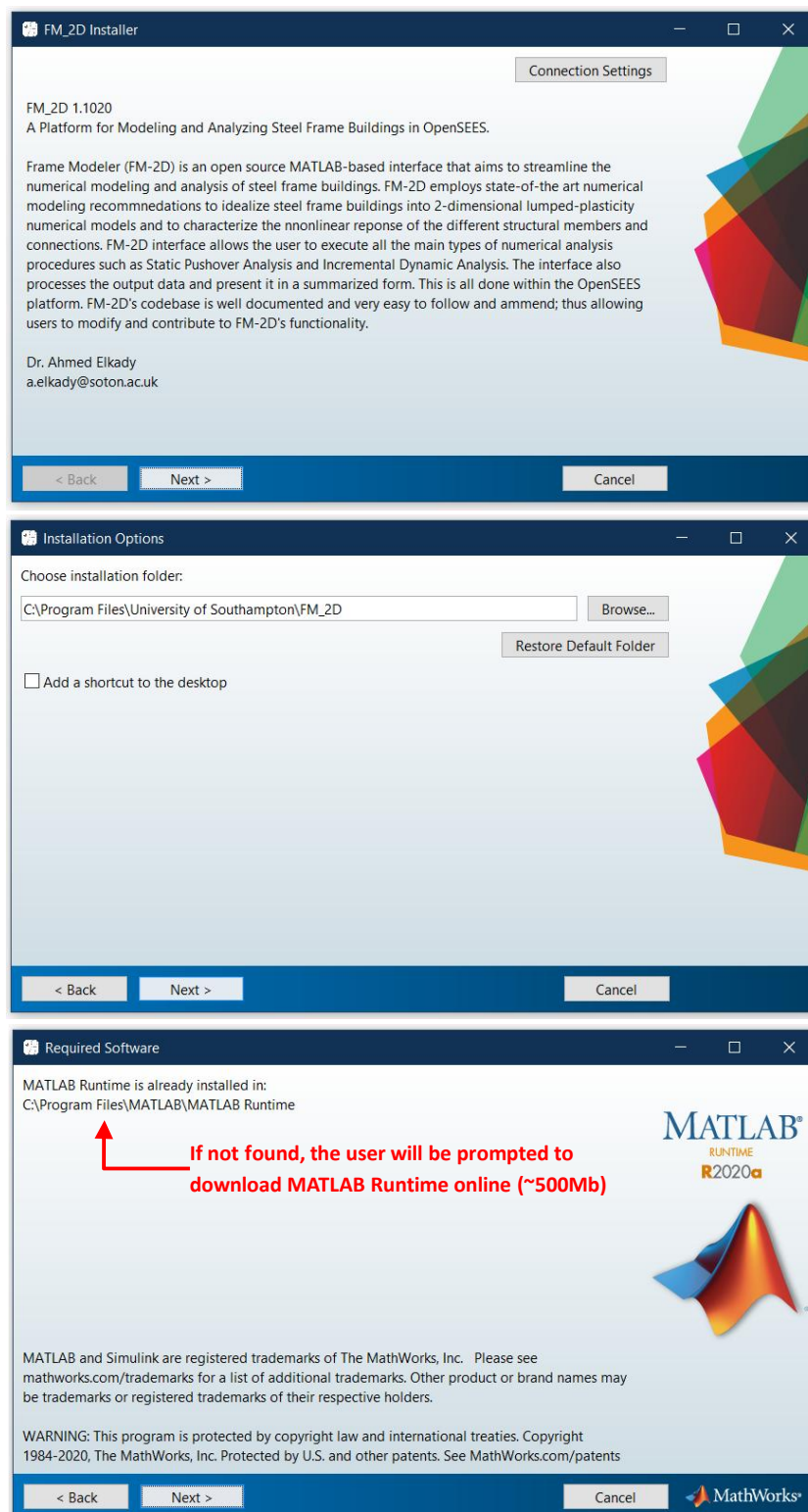


Figure 1.2. Installation steps

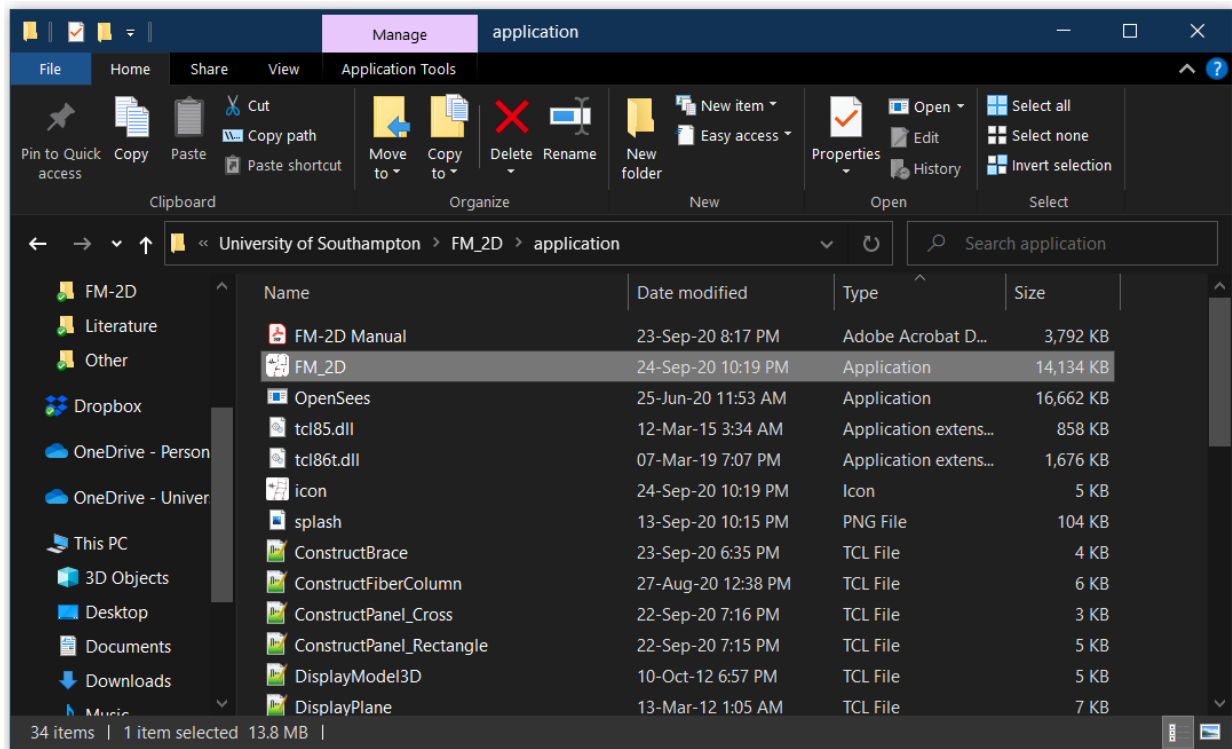


Figure 1.3. Installed files and folders in the default installation directory

2. Introduction

The performance-based earthquake engineering (PBEE) framework (SEAOC 1995; Cornell and Krawinkler 2000; FEMA 2000; Porter 2003; Deierlein 2004; Krawinkler and Miranda 2004; Moehle and Deierlein 2004; Aslani and Miranda 2005; Yang et al. 2009; FEMA 2012a; Hamburger et al. 2012) has been increasingly used in designing and retrofitting buildings, particularly within the past decade. At its core, this framework relies on the availability of robust response data for a building under a given earthquake scenario in order to be able to reliably assess/quantify structural performance, in terms of damage and economic losses, beyond the scope of conventional building codes. Structural response data includes global engineering demand parameters (EDP) such as the story-drift ratios (*SDR*), the residual drift ratios (*RDR*) and peak floor accelerations (*PFA*). Local EDP includes for instance connection and member rotational demands. Obtaining structural response data is arguably the most demanding and time-consuming step of the PBEE framework. This step involves 1) numerical modeling the building in a structural analysis platform, 2) determine the nonlinear hysteretic response parameters of the different connections and members, 3) defining the loads and masses, 4) conducting static and dynamic analysis procedures using a suite of ground motion records, and finally, 5) reading and processing response data. This is laborious step can take weeks and sometime months to complete. Most importantly, this step requires a skilled and experienced user who knows the specifics of state-of-the-art numerical modeling guidelines and simulation techniques. While existing software packages (Ramirez and Miranda 2009; FEMA 2012b; SP3 2017; Elkady et al. 2018), for economic loss assessments, provide assistance with most of the PBEE framework steps (such as the damage fragility data of structural component and contents and the associated consequences), the burden of obtaining structural response data remain entirely on the user. Some software provides tools to generate structural response data based on simplified methods. As such, the accuracy of the generated data can be low and the associated uncertainty in the corresponding computed economic losses can be large.

To that end, an open-source MATLAB-based (MATLAB 2019) standalone software is developed and presented herein. The software is named Frame Modeler-2D (FM-2D) and it aims to streamline and speed the numerical modeling and analysis of steel frame buildings. Using the Open System for Earthquake Engineering Simulation Platform (*OpenSees*) (Mckenna 1997; Mazzoni et al. 2006), FM-2D builds robust 2-dimensional (2D) numerical models of steel frame buildings using state-of-the-art numerical modeling guidelines, conducts different types of nonlinear analysis procedures and processes and reports the structural response data in a comprehensive manner. FM-2D has a well-structured and documented codebase; thus, allowing it to be a platform for engineers and researchers to further develop and contribute to its modeling and simulation capabilities.

3. Operational Outline

FM-2D's operational outline is illustrated in Figure 3.1. A project definition follows a sequential three-step procedure. In brief, the procedure involves: 1) selecting the frame type and the basic numerical model parameters, 2) defining the building geometry, member sizes, loads, and materials, and 3) defining the analysis type and its parameters. Using the user-defined data, FM-2D constructs and generates *tcl* files of the numerical model, executes the analysis in *OpenSees*, processes the analysis data, and stores structural response in *txt* files for the user to visualize or to use externally with other software packages. The different graphical user interface (GUI) modules in FM-2D that are used to define the model, building, and analysis data are discussed in detail in Section 4. The technical details for generating the 2D numerical models are discussed in Section 7.

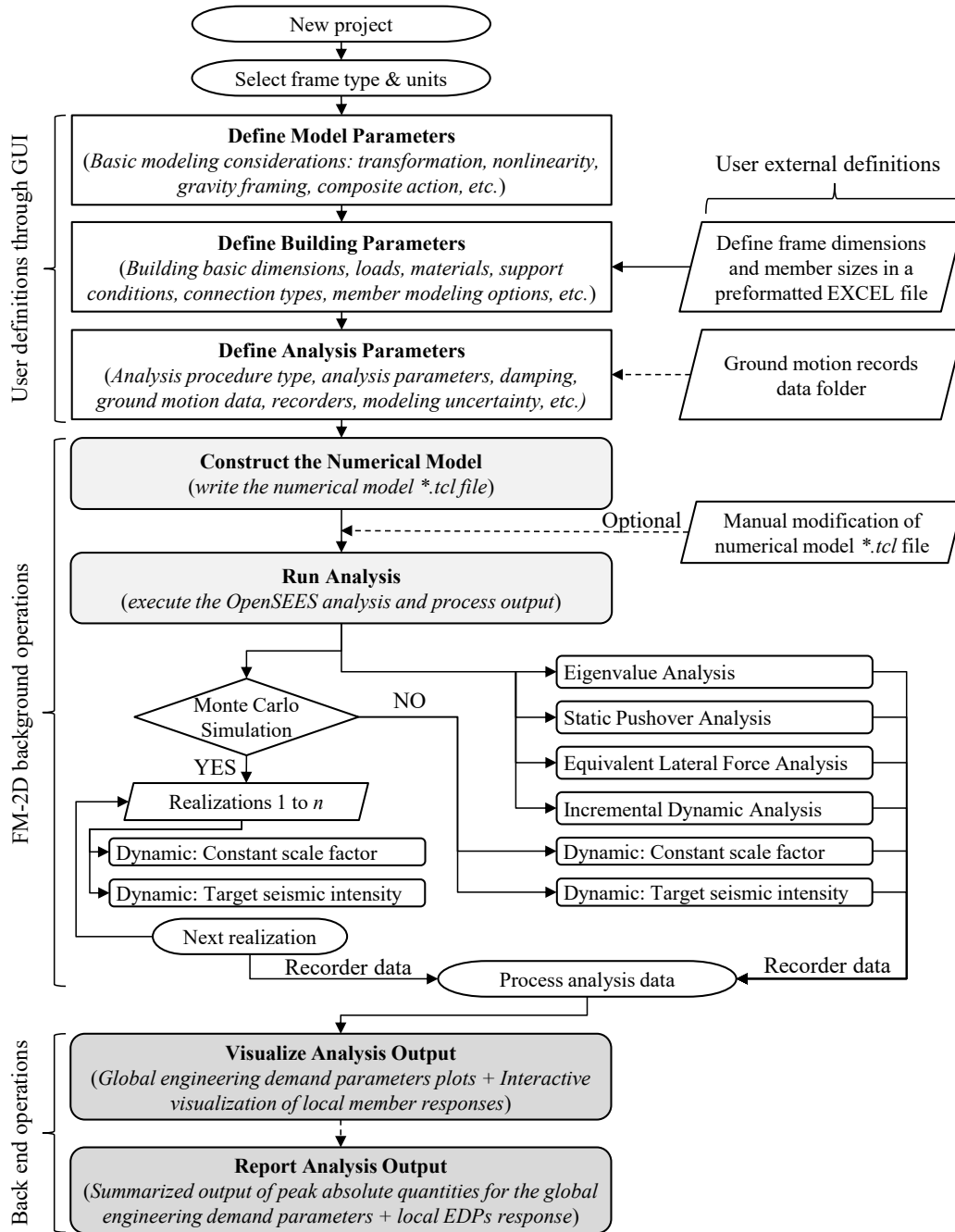


Figure 3.1. FM-2D operational outline

4. Interface Components

4.1 Main Console

Figure 4.1 illustrates FM-2D's main console, which is divided into three main panels:

- 5) Panel A is the project file management panel, which includes the following buttons:
 - “New” create a new project file
 - “Open” open an existing project file
 - “Save As” save an open project under a different name
 - “Scope” shows a summary of the current project definitions
- 6) Panel B includes three buttons that define the three main components of any project:
 - “Model Parameters” defines the main numerical modeling parameters
 - “Building Parameters” defines the building geometry and numerical modeling details
 - “Analysis Parameters” defines the type of analysis and its parameters
- 7) Panel C includes the execution buttons:
 - “Get period” for probing the building's first-mode period, seismic weight, and mass.
 - “Build tcl file” for building a standalone *tcl* file of the building model.
 - “Run Model” for running the numerical analysis.
 - “Visualize” for visualizing the numerical analysis results.

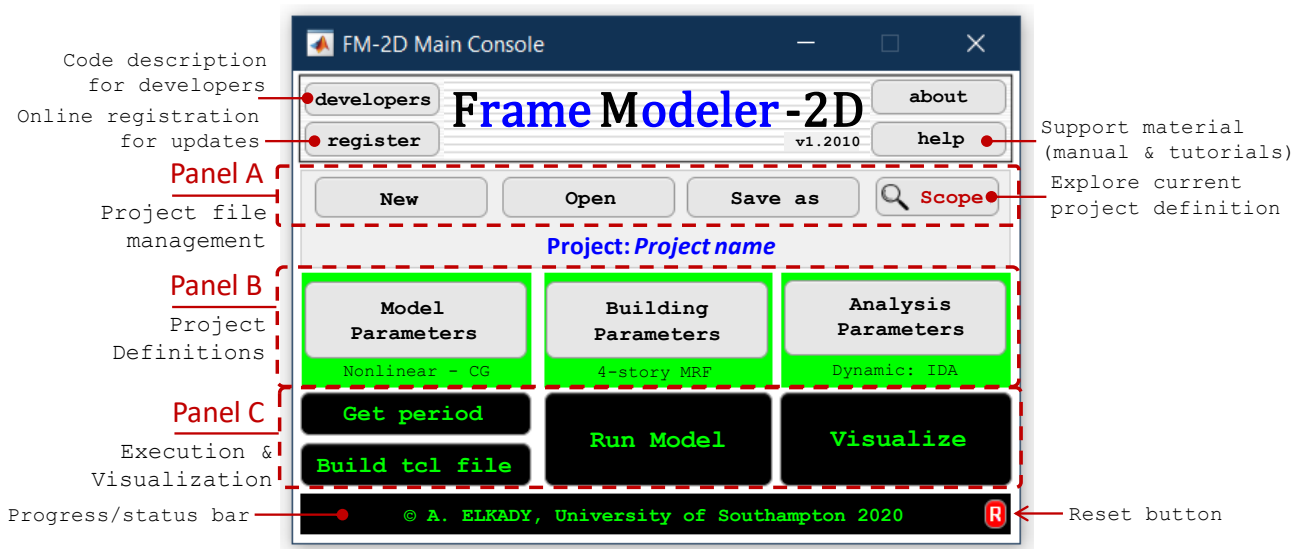


Figure 4.1. FM-2D v4.2605 main console

The FM-2D project is saved as a transferable MATLAB *.mat file and can be loaded through the toolbox at any time from different machines. When starting a new project, the user is prompted to select the building frame type (MRF or CBF) and the project units (“kN & mm” or “kip & in”) as shown in Figure 4.2. Interface for selecting the frame type and project units. The user can also provide an optional description of the project for reference.

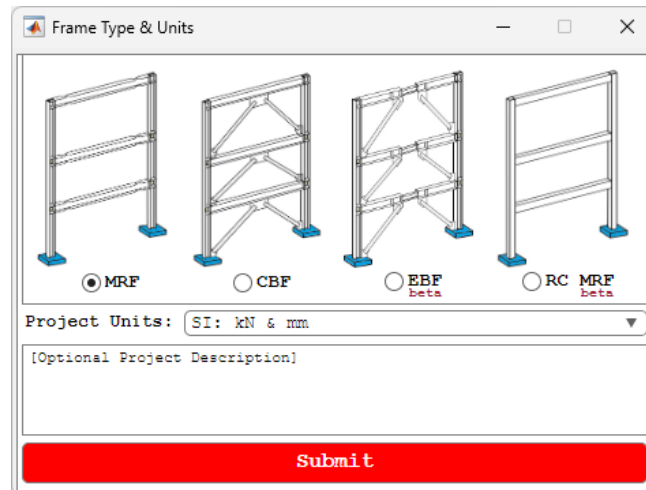


Figure 4.2. Interface for selecting the frame type and project units

4.2 Model Parameters

As demonstrated in Figure 4.3, FM-2D idealizes a given steel frame building, and its response in a given orthogonal direction, as a 2-dimensional numerical model. Particularly, the idealized 2D model comprises one of the main lateral-force-resisting planar frames in the considered orthogonal direction (noted hereafter simply as MF). In FM-2D v1.2010, this main frame (MF) can be either a moment-resisting frame (MRF) or a concentrically braced frame (CBF). The rest of the building, that is the gravity framing system (GFS) tributary to the considered MF, is represented by a single-bay fictitious equivalent gravity frame (EGF) as proposed by Gupta and Krawinkler (1999) and discussed in Lignos et al. (2011), Lignos et al. (2013) and Elkady and Lignos (2015). The EGF captures the strength and stiffness of the gravity framing system (if chosen to be considered by the user) and most importantly, it is used to represent the gravity P-Delta loads of the building in 2D. The EGF is connected to the MF using rigid or flexible truss floor links, that are meant to transfer the floor inertia forces between the two systems. The 2D numerical model uses mainly the lumped plasticity modeling approach (also known as point plasticity or concentrated plasticity) to idealize the structural members and connections behaviors. This is discussed in detail in Section 7.

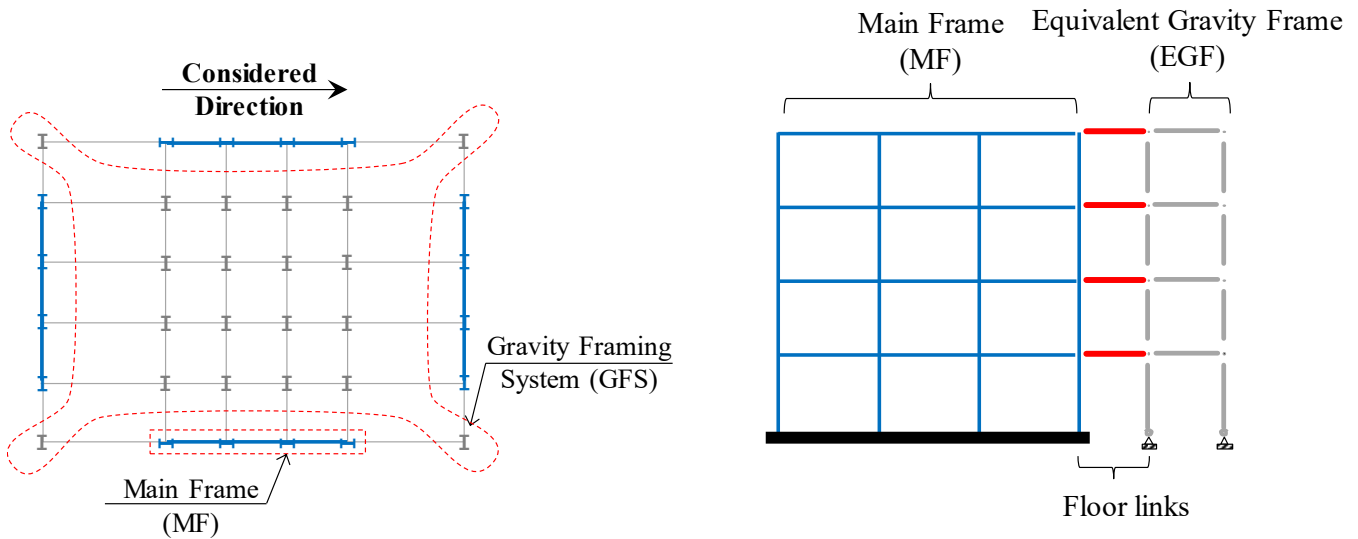


Figure 4.3. Building numerical representation in 2D

The first step in an FM-2D project definition is to specify the “Model Parameters”. When the “Model Parameters” button is clicked, the user is prompted to the module shown in Figure 4.4, regardless of the selected frame type. The module contains two tabs, described as follows:

4.2.1 Modeling Options tab

Geometric Transformation: from the dropdown menu, select the type of geometric transformation that will be assigned to the beam and column elements in your model: Linear, P-delta, or Correlational. Refer to *OpenSees* wiki for more details: https://opensees.berkeley.edu/wiki/index.php/Geometric_Transformation_Command

Numerical Model Type: from the dropdown menu, select whether the numerical model would be Linear or Nonlinear. The Linear option can be used to create an elastic linear model, which can be used for instance to conduct an equivalent lateral force analysis as per ASCE (2016).

Panel Zone Dimension & Deformation: Mark the checkbox to consider the column web panel zone –rectangular– dimension and its rotational flexibility (i.e., shear distortion deformation) under shear loads. If the check box is not marked, the panel zone flexibility is not considered, where it will be idealized as a rigid cruciform shape as discussed in detail in section 7.4 (refer to Figure 7.2).

Rigid Floor Diaphragm: Mark the checkbox to tie all nodes at each floor level with a diaphragm constraint (i.e., all nodes have equal lateral degrees of freedom).

Gravity Framing System: Mark the checkbox to consider the strength and stiffness of the gravity framing system’s columns, beams, and beam-to-column shear connections, as per Elkady and Lignos (2015). The parameters of the GFS are defined later in the Building module.

Composite Action of the Floor Slab: Mark the checkbox to consider the effects of the composite action provided by the floor composite –concrete on steel deck– slab when modeling the MF and GFS beams and beam-to-column connections, as per the recommendations of Elkady and Lignos (2014, 2015). When considered, the user shall specify the amplification factors for the bare steel elastic stiffness ratio of the MF and gravity beams. This will be specified within the Building module in the next step.

4.2.2 Member Modeling tab

The “Member Modeling” tab is shown in Figure 4.4(b). The modeling parameters to be defined are as follows:

MF Column Elements: specify an option for modeling the MF column and beam elements. Three options are available:

- 1) *elasticBeamColumn* elements and IMK rotational springs at the ends.
- 2) *dispBeamColumn* elements with fiber section and the updated Voce-Chaboche (UVC) material model.
- 3) *forceBeamColumn* elements with fiber section and the updated Voce-Chaboche (UVC) material model.

If option 2 or 3 is selected, specify the element modeling parameters, including: 1) the number of segments along the element length, 2) the number of integration points per segment, and 3) the global mid-length geometric imperfection (i.e, amplitude of sin wave-shaped imperfection) as a multiple of the element length.

Account for shear flexibility: since elastic beam-column elements are based on Bernoulli’s beam theory, they do not capture shear deformations. To supplement that, the user can select this option to apply a reduction factor to the beam’s moment of inertia to account for the shear deformations.

MF Column P-M Interaction: specify the method that will be used to obtain the axial load level P that will in turn be used to reduce the main frame’s columns’ plastic flexural strength, M_y . Two methods are available:

- 1) Compute P based on the gravity load (i.e., the seismic weight) multiplied by an optional scale factor. For a column at story i in an n -story building, P is computed using Equation 4.7 in which P_t is the sum of column tributary loads in the floors above the column.

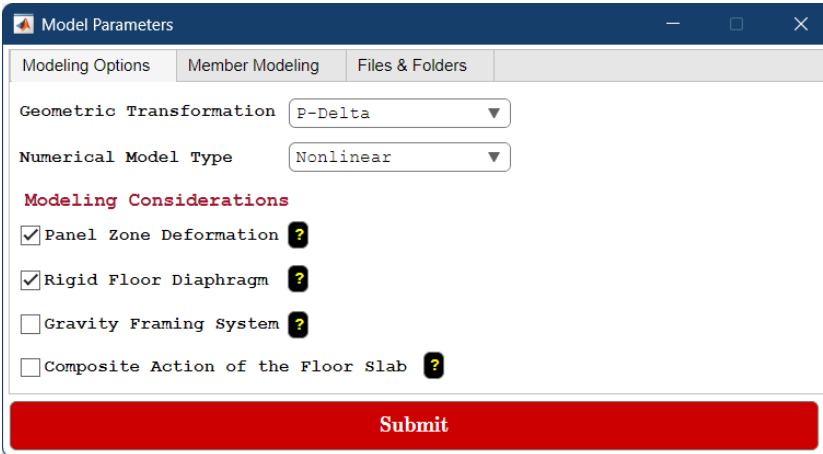
$$4) \quad P = \text{scale factor} \cdot P_t = \text{scale factor} \cdot \sum_{\text{floor}=i+1}^{\text{floor}=n+1} L_t \quad (4.1)$$

- 2) Compute P based on the maximum compressive axial load level to be experienced by the column during a pushover analysis, $P_{\text{PO,max}}$, as follows:

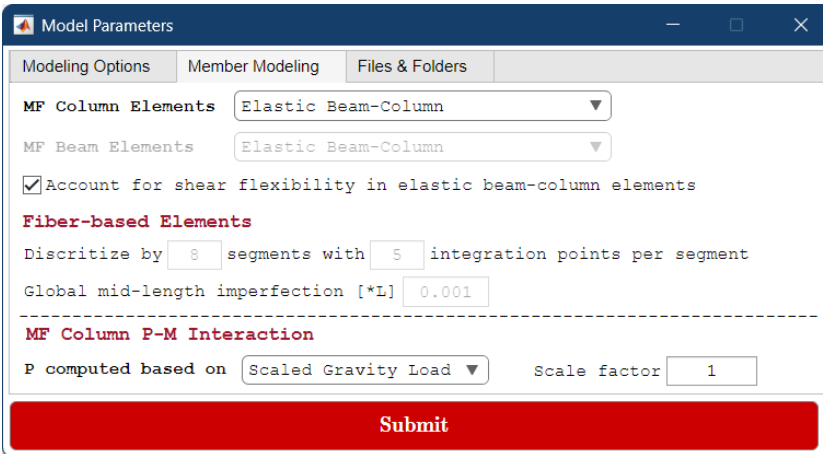
$$5) \quad P = P_t + (P_{\text{PO,max}} - P_t) / 2 \quad (4.2)$$

Note that method #2 will require FM-2D to run a preliminary Pushover analysis prior to the main analysis.

(a)



(b)



(c)

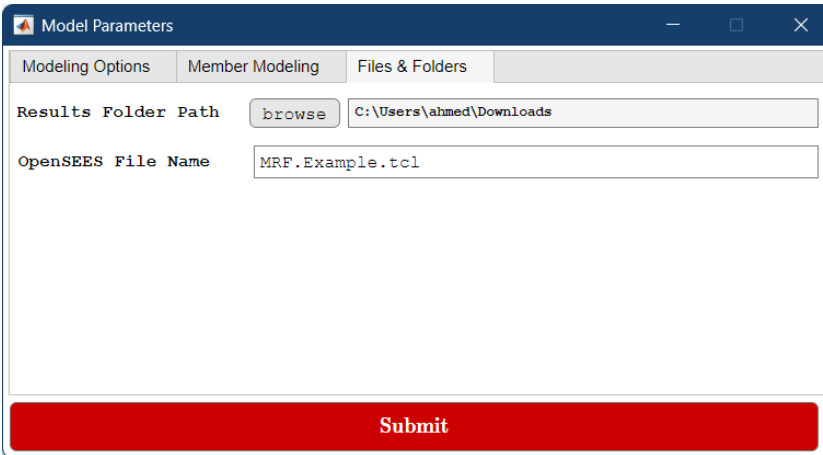


Figure 4.4. Model parameters module

4.2.3 Files & Folders tab

Results Folder Name: using the “browse” button, the user can select a directory to save the analysis results. By default, this directory is taken the same as the project file directory.

OpenSees File Name: specify in the textbox the name of the numerical model *tcl* file that will be created by FM-2D (e.g., *TestFrame.tcl*). Note that this name should not include any spaces or special characters. By default, this filename is taken the same as the project name minus the spaces (if any).

4.3 Building Parameters

The second step is to define the “Building Parameters”. When the “Building Parameters” button is clicked, the user is prompt to a module containing four tabs, described as follows:

4.3.1 Building Geometry tab

To aid the subsequent discussion and description of geometric parameters, Figure 4.5 shows the plan layout of a generic building with its basic components annotated. In this figure, the two orthogonal directions are labeled “Direction 1” and “Direction 2”. The former is assumed here to be the “considered” direction for the numerical model/analysis to be conducted. The “Building geometry” tab is shown in Figure 4.6. In this tab, the user shall specify some basic geometric and configuration parameters including:

- 1) the building dimensions in the plan layout.
- 2) the floor area tributary to the MF’s exterior and interior columns as demonstrated in Figure 4.5.
- 3) the number of MFs in the considered direction (e.g., two MFs in the generic building in Figure 4.5).
- 4) the MF’s number of stories and bays. Note that for CBFs, FM-2D v2.2505 currently allows only single-bay CBFs.
- 5) the GFS’s number of columns and beams. Note that the number of GFS columns should be that in the entire building layout (i.e., 20 columns in the generic building) and 2) the number of GFS beams should be that in the considered orthogonal direction only (i.e., 24 beams in the generic building).

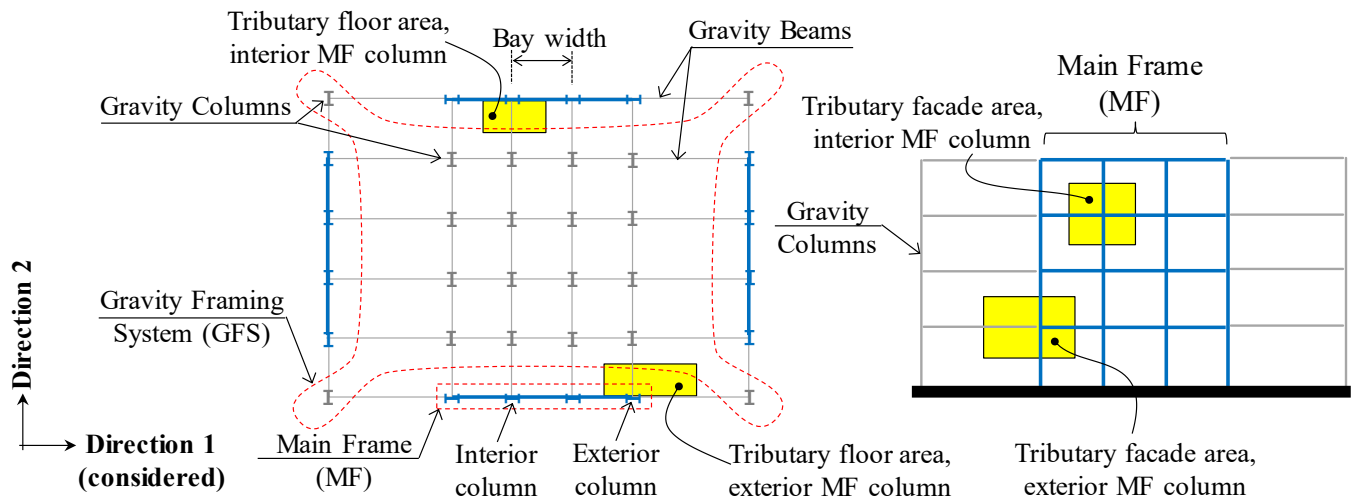


Figure 4.5. Components of a generic regular building plan and elevation layout

Finally, the user shall define building story heights, bay widths and member sizes. This is done in a pre-formatted EXCEL file that is provided within FM-2D supporting documents (see *docs* folder in GitHub). The EXCEL file includes several sheets as follows:

For MRFs, CBFs, and EBFs:

W_bay: specify the width of the MF bays (using the same units of the projects).

H_story: specify the heights of the MF stories (using the same units of the projects) as demonstrated in Figure 4.7a.

MF Columns: specify the cross-sections of the MF columns at each story and axis as demonstrated in Figure 4.7b. Note that the user can choose any cross-section of those included in the integrated database, using the same cross-section notation. Also note that if there is a splice at a given story i (i.e., change in column size), the user shall specify –for that story i – the column cross-section size of the lower story $i-1$. For other sections that are not in the integrated section database, the user can define custom I-shaped sections by entering the following “BU $dx b_f x t_w x t_f$ ” where BU stands for built-up, d is the section depth, b_f is the section flange width, t_w is the web thickness, and t_f is the flange thickness. For instance: BU 500x200x12x15. Note that these section dimensions should be in the same units as the defined project.

MF Beams: specify the cross-sections of the MF beams at each floor and bay as demonstrated in Figure 4.7c. Note that the user can choose any cross-section of those included in the integrated database, using the same cross-section notation. For other sections that are not in the integrated section database, the user can define custom I-shaped sections by entering the following “BU $dx b_f x t_w x t_f$ ” where BU stands for built-up, d is the section depth, b_f is the section flange width, t_w is the web thickness, and t_f is the flange thickness. For instance: BU 500x200x12x15. Note that these section dimensions should be in the same units as the defined project.

GFS Columns: specify the cross-sections of the GFS columns at each story. Note that the user can choose any cross-section of those included in the integrated database, using the same cross-section notation. Also, note that GFS columns are assumed to be the same at all axes.

GFS Beams: specify the cross-sections of the GFS beams at each floor. Note that the user can choose any cross-section of those included in the integrated database, using the same cross-section notation. Also, note that GFS beams are assumed to be the same in all bays.

Column Splice: specify the presence of a splice in each story. Specify “0” for no splice and “1” for splice, as demonstrated in Figure 4.7d. The user can also specify the relative location of the splice within the respective story height, relative to the bottom floor. By default, the splices are allocated at mid-height of the story (i.e., relative location = 0.5). Also note that the splice connection is assumed, in the current release, to be fully rigid.

Doubler Plates: specify the total thickness of the doubler plates in the column web panel zone region at each floor and axis (if any) as demonstrated in Figure 4.7e.

Fs Profile: specify the force coefficients for the floor links at each floor as demonstrated in Figure 4.7f. These coefficients will be multiplied by the reference force, F_s , specified to under the “Supports & Connections” tab, to obtain the “limiting force” that will be assigned to the floor truss link. Note that these coefficients are set equal to 1.0 by default if the F_s profile is assumed constant. These coefficients need only be specified if a non-uniform F_s profile is selected.

For MRFs only:

MF EEP: if the selected connection type is a semi-rigid extended endplate, the user shall define the parameters describing the connection layout. The connection shall be described using the following notation,

$$tep \times bep \times d_b \times pt \times g \times nrows \times StiffenerC$$

where, *tep* is the endplate thickness, *bep* is the endplate width, *d_b* is the bolt nominal diameter, *pt* is the distance between the bolt rows above and below the beam flange, *g* is the bolt gauge distance (i.e., the distance between the bolt columns), *nrows* is the number of rows in tension (2 or 3), *StiffenerC* is the presence of column stiffeners/continuity plates (0 or 1).

GFS Connection: if the GFS is considered, the user shall define the parameters describing the GFS connection layout. For semi-rigid extended endplate connections, the definition is as discussed previously. For flush endplate connections, the connection shall be described using the following notation,

$$tep \times d_b \times dt \times g \times nrows \times StiffenerC$$

where, *tep* is the endplate thickness, *d_b* is the bolt nominal diameter, *dt* is the distance between the tension bolt rows and middle of the beam tension flange, *g* is the bolt gauge distance (i.e., the distance between the bolt columns), *nrows* is the number of rows (2 or 3), *StiffenerC* is the presence of column stiffeners/continuity plates (0 or 1).

For CBFs and EBFs only:

Braces: specify the cross-sections of the braces at each story as demonstrated in Figure 4.7g. Note that the user can choose any cross-section of those included in the integrated database, using the same cross-section notation.

CGP t_{plate}: specify the thickness of the corner gusset plate (CGP) at each story as demonstrated in Figure 4.7h.

MGP t_{plate}: specify the thickness of the mid-span gusset plate (MGP) at each story.

CGP L₁₂₃: specify the characteristic lengths L_1 , L_2 , and L_3 of the corner gusset plate (CGP) at each story as demonstrated in Figure 4.8 (refer to Hsiao et al. (2013) details about these characteristic lengths).

MGP L₁₂₃: specify the characteristic lengths L_1 , L_2 , and L_3 of the mid-span gusset plate (MGP) at each story.

CGP L_{offset}: specify the offset length between the brace end and beam-to-column centerlines intersection, for the corner gusset plate (CGP) at each story (refer to Figure 4.8).

MGP L_{offset}: specify the offset length between the brace end and beam-to-column centerlines intersection, for the mid-span gusset plate (MGP) at each story (refer to Figure 4.8).

CGP L_connection: specify the length of the connection between the brace and the corner gusset plate (CGP) at each story (refer to Figure 4.8) as demonstrated in Figure 4.7j.

MGP L_connection: specify the length of the connection between the brace and the mid-span gusset plate (MGP) at each story (refer to Figure 4.8).

MGP W_plate: specify the width of the mid-span gusset plate (MGP) at each story (refer to Figure 4.8).

L_brace: specify the end-to-end length of the brace at each story (refer to Figure 4.8).

For EBFs only:

MF_SL: specify the cross-sections of shear links at each floor.

For RC MRFs:

The sections for the RC frame's beams and columns are defined in the Excel sheet using the following notation (see template Excel sheet *RCMF_2Story.xlsx* for example):

For rectangular cross-sections:

$H \times B \times coverH \times coverB \times nBarTop \times dBarTop \times nBarBot \times dBarBot \times nBarInt \times dBarInt \times nBarShear \times dBarShear \times s$

For circular cross sections:

$D \times cover \times nBar \times dBar \times nBarShear \times dBarShear \times s$

where, H and B are the section's depth and width, respectively, D is the section's diameter, $coverH$ and $coverB$ are the distances from the section boundary to the neutral axis of reinforcement in the vertical and horizontal direction, respectively, $cover$ is the distance from the section boundary to the neutral axis of the reinforcement, $nBarTop$, $nBarBot$ and $nBarInt$ are the number of reinforcing bars in the top layer, bottom layer, and intermediate layer, respectively, $dBarTop$, $dBarBot$ and $dBarInt$ are the diameters of the reinforcing bars in the top layer, bottom layer, and intermediate layer, respectively, $nBarShear$ are the number of shear reinforcing bars (i.e., legs of the stirrup), $dBarShear$ is the diameter of the reinforcing bars in shear, s is the shear reinforcement (stirrup) spacing, $\# \rho_T$ is the area ratio of longitudinal bottom reinforcement (tension), $\# \rho_C$ is the area ratio of longitudinal top reinforcement (compression), $\# \rho_I$ is the area ratio of longitudinal middle reinforcement, $\# \rho_{sh}$ is the area ratio of transverse reinforcement in the plastic hinge region spacing

Figure 4.6. Building parameters module: Building geometry tab

	A	B	C
1		Height	
2	Story 1	180	
3	Story 2	156	
4	Story 3	156	
5	Story 4	156	
6	Story 5		

Ready

(a)

	A	B	C	D	E	F
1		Axis 1	Axis 2	Axis 3	Axis 4	Axis
2	Story 1	W24X103	W24X103	W24X103	W24X103	
3	Story 2	W24X103	W24X103	W24X103	W24X103	
4	Story 3	W24X103	W24X103	W24X103	W24X103	
5	Story 4	W24X62	W24X62	W24X62	W24X62	
6	Story 5					

Ready

(b)

	A	B	C	D	E
1		Bay 1	Bay 2	Bay 3	Bay 4
2	Floor 2	W21X73	W21X73	W21X73	
3	Floor 3	W21X73	W21X73	W21X73	
4	Floor 4	W21X57	W21X57	W21X57	
5	Floor 5	W21X57	W21X57	W21X57	
6	Floor 6				

Ready

(c)

	A	B	C
1		Splice	Relative Location
2	Story 1	0	0.5
3	Story 2	0	0.5
4	Story 3	1	0.5
5	Story 4	0	0.5
6	Story 5		

Ready

(d)

	A	B	C	D	E	F
1		Bay 1	Bay 2	Bay 3	Bay 4	Bay 5
2	Floor 2	0	0.3125	0.3125	0	
3	Floor 3	0	0.3125	0.3125	0	
4	Floor 4	0	0.3125	0.3125	0	
5	Floor 5	0	0.3125	0.3125	0	
6	Floor 6					

Ready

(e)

	A	B	C
1		Fs Coeff.	
2	Floor 2	1	
3	Floor 3	1	
4	Floor 4	1	
5	Floor 5	1	
6	Floor 6		

Ready

(f)

	A	B	C
1		Bay 1	Bay 2
2	Story 1	HSS12.5X.500	
3	Story 2	HSS12.5X.500	
4	Story 3	HSS11.25X.500	
5	Story 4	HSS9.625X.500	
6	Story 5	HSS9.625X.375	
7	Story 6	HSS7.5X.312	
8	Story 7		

Ready

(g)

	A	B	C
1		Bay 1	
2	Story 1	0.625	
3	Story 2	0.625	
4	Story 3	0.625	
5	Story 4	0.625	
6	Story 5	0.5	
7	Story 6	0.5	
8	Story 7		

Ready

(h)

	A	B	C	D
1		L1	L2	L3
2	Story 1	11.8722	25.579	3.7405
3	Story 2	6.5786	23.3259	1.4874
4	Story 3	7.5914	10.9283	30.4097
5	Story 4	7.6514	18.0122	1.0753
6	Story 5	7.0529	14.5269	0.4768
7	Story 6	7.185	13.4558	3.9323
8	Story 7			

Ready

(i)

	A	B	C
1		Bay 1	Bay 2
2	Story 1	27	
3	Story 2	27	
4	Story 3	24	
5	Story 4	21	
6	Story 5	26	
7	Story 6	10	
8	Story 7		

Ready

(j)

Figure 4.7. Dimensions and member sizes definitions in the pre-formatted EXCEL file

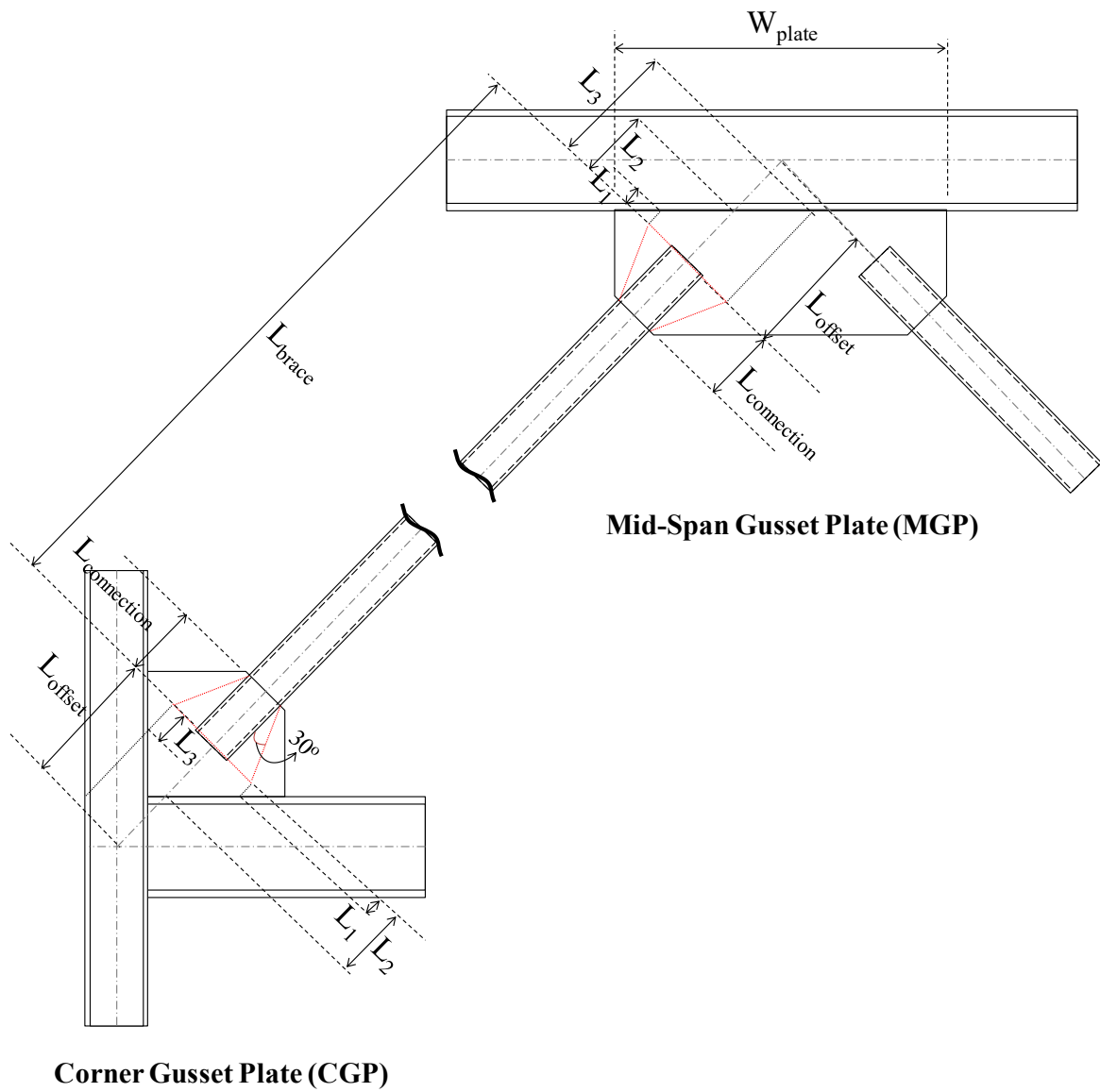


Figure 4.8. Geometric parameters used to define the CBF brace-to-gusset connection

4.3.2 Support & Connections tab

The “Support & Connections” tab is shown in Figure 4.9. In these tabs, the user shall specify the support conditions and connection types. In particular:

MF column supports: select, from the dropdown menu, whether the main frame columns have a fixed or pinned support at the ground floor level.

MF column splice condition: specify whether the column splices are shear or moment splices. For the shear option, the joint is assigned a zero rotational spring while for the moment option, the joint is assigned an elastic-rigid rotational spring the dropdown menu, whether the main frame columns have a fixed or pinned support at the ground floor level.

MF beam-to-column connection:

- For moment frames, select from the dropdown menu the main frame beam-to-column connection type. For MRFs, two types are available: 1) reduced beam section connection (RBS) and 2) Welded/Bolted Full Section (i.e., other-than-RBS connection as per Lignos and Krawinkler (2011)). In the case of the former, the user shall specify the additional geometric parameters a , b , and c as demonstrated in Figure 4.9. Further details about the connections geometries can be found in AISC (2016).
- For braced frames, three types of connection are available to choose from: 1) Pinned (i.e., zero stiffness and strength), 2) Shear (i.e., a shear-tab connection with pre-defined pinched response as per Elkady and Lignos (2015)), or 3) Moment (i.e., fully rigid) connection. For CBFs, the connection type can be for odd- and even-numbered floors, which can be particularly useful for X-bracing configurations where no gusset plate connections are present at odd-numbered floor levels.

MF beam lateral bracing: specify the spacing at which the beam is laterally braced.

MF beam mid-span condition: for braced frames, specify whether the mid-span point (at the mid-span gusset plate connection) is allowed to sag following brace buckling.

GFS specifics: When the GFS is considered, the user shall also specify (using the dropdown menu) the orientation of the gravity columns cross-section axis with respect to the considered orthogonal building direction, the base support condition for the GFS columns, and the type of the GFS beam-to-column connection.

MF to EGF link condition: select, from the dropdown menu, the type of the MF-to-EGF floor link whether it is a rigid or flexible link. The floor link is meant to represent the flexibility of the floor diaphragm to the main frame connection. In FM-2D, this link is represented by truss members with an infinitely stiff bilinear axial force-axial deformation response as demonstrated in Figure 4.10. The bilinear response is defined using the break force, F_s , and the post-break force stiffness, K_s . Typically, and by default in FM-2D, a rigid link is considered at each floor level; thus, a sufficiently large break force is assumed. For the flexible link option, the user shall specify the values for F_s and K_s . Additionally, the user can specify if the F_s force varies at different floors. In case it does, the F_s force coefficients need to be specified in the pre-formatted EXCEL as discussed in the previous section.

Building Parameters

Building Geometry | **Supports & Connections** | Loads | Materials | Other

MF Column Supports: Fixed

MF Column Splice Condition: Fixed

MF Connection: Reduced Beam Section

$a = 0.625$ b_f
 $b = 0.750$ d
 $c = 0.250$ b_f

MF Beam is laterally braced at 0.500 L segments

Column orientation: Weak-axis
 Support: Pinned
 Gravity connection type: Shear-tab

MF to EGF Link Condition: Rigid

☐ Non-uniform F_s Profile
 $F_s = 1e+04$ kN
 $K_s = 0$ kN/mm

Submit

Figure 4.9. Building parameters module: Supports and connections tab

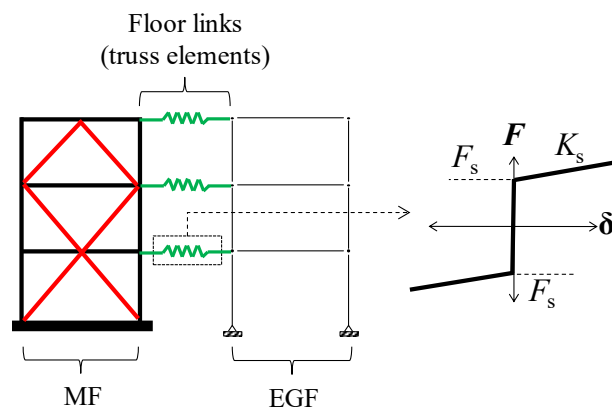


Figure 4.10. Floor truss links; MF-to-EGF connection

4.3.3 Loads tab

Loads: specify the uniform area loads for the pre-defined load cases: Dead load (DL), Live load (LL), Generic load (GL), and Cladding load (CL). For DL, LL, and GL, loads can be specified for the typical floor and roof levels.

Load Combination Coefficients: specify the coefficients of the load combinations that will be used to compute the seismic weights, W_s , and the building mass, M . For a specific floor level, W_s and M are computed as follows,

$$6) \quad W_s = C_{w,DL} \cdot DL + C_{w,CL} \cdot CL + C_{w,LL} \cdot LL + C_{w,GL} \cdot GL \quad (4.3)$$

$$7) \quad M = C_{m,DL} \cdot DL + C_{m,CL} \cdot CL + C_{m,LL} \cdot LL + C_{m,GL} \cdot GL \quad (4.4)$$

In which, C_w and C_m parameters are the loading combination coefficients for the seismic weight and mass computation, respectively. In the 2D model, the gravity loads and masses are computed and applied at nodal point at each floor level as demonstrated in Figure 4.12. In particular, the MF's columns are assigned loads and masses corresponding to their tributary area; L_t and m_t , respectively which are computed as follows:

$$8) \quad L_t = A_{trib,Column} \cdot W_s \quad (4.5)$$

$$9) \quad m_t = A_{trib,Column} \cdot M \quad (4.6)$$

Note here that $A_{trib,Column}$ refers to the MF column's tributary "floor" area for DL, LL and GL while it refers to the tributary "façade" area for CL, as demonstrated in Figure 4.5. The rest of the building load and mass (L_g and m_g , respectively) is assigned to the EGF grid nodes as demonstrated in Figure 4.12. These are computed as follows:

$$10) \quad L_g = (A_{trib,MF} - A_{trib,Column}) \cdot W_s \quad (4.7)$$

$$11) \quad m_g = (A_{trib,MF} - A_{trib,Column}) \cdot M \quad (4.8)$$

Where, $A_{trib,MF}$ refers to the "floor" or "façade" area tributary to one MF in the considered direction. For the sample building shown in Figure 4.12, this tributary floor area would be equal to half of the floor area (that is the total floor area divided by two MFs that are available in the considered direction).

Load application: specify whether the gravity loads are applied as point loads at the column ends or as distributed uniform loads on the beams.

Building Parameters

Building Geometry

Supports & Connections

Loads

Member Modeling

Materials

Loads [kN/m²]

	Typical	Roof
Floor Dead Load	4.31	4.31
Floor Live Load	2.4	0.96
Cladding Load	1.2	
Floor Generic Load	0	0

Load Combination Coefficients

☒ default values

	Dead	Live	Cladding	Generic
Seismic weight	1.05	0.25	1.05	1.00
Mass	1.00	0.00	1.00	1.00

Load application option

Column point load

Submit

Figure 4.11. Building parameters module: Loads tab

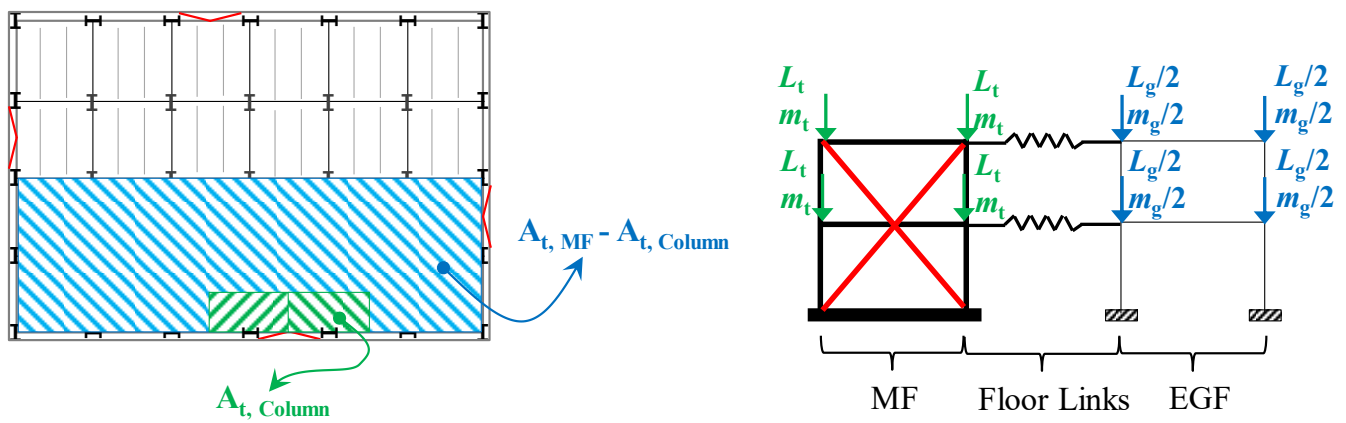


Figure 4.12. Illustration of mass and gravity load computation and allocation in the numerical model

4.3.4 Material tab

Specify the material properties for the different structural components, in the same unit as that of the project. This includes the modulus of elasticity (E), the yield stress (f_y), and Poisson's ratio (μ). Note that f_y values should be the nominal ones and not the measured ones, since FM-2D assigns amplification factors (to get the expected yield value) internally for the different components as part of the nonlinear spring subroutines. If fiber-based models are being used for either the column or the beam, the user can specify the expected-to-nominal yield stress ratio (R_y) for the column and the beam.

Using the dropdown menu, the user can also select a steel type/grade from an integrated material library. This material library and associated properties are based on extensive material calibrations done by de Castro e Sousa and Lignos (2018); de Castro e Sousa et al. (2020) against the Voce-Chaboche constitutive plasticity model (von Mises 1913; Voce 1948; Chaboche 1989). Check this *OpenSees* wiki [link](#) for more details.

Material [kN/mm ²]	E	f_y	f_u	μ
Beams & columns	200	0.355	0.47	0.3
Brace		55		
Gusset plate		55		
Endplate		0.35	0.47	
Bolt		0.9	1	
Rebar	2.9e+04	55		0.3
Concrete	2600	5		0.2

	Column	Beam
Expected to nominal stress ratio (R_y)	1.15	1.17

Submit

Figure 4.13. Building parameters module: Materials tab

4.3.5 Other tab

Specify the modeling parameters for the composite action. This includes the floor slab dimensions: the steel deck rib height (t_{rib}) and the thickness of the concrete slab above the steel deck rib (t_{slab}). By default, those are taken as 3.5 inches and 4.0 inches, respectively. The stiffness coefficients that represent the amplification in the MF or the GF beams' flexural stiffness due to the composite action. These factors are taken equal to 1.4 by default (Elkady and Lignos 2014).

4.4 Analysis Parameters

The “Analysis Parameters” module is shown in Figure 4.14. This module contains six tabs as follows:

4.4.1 Analysis Options tab

Analysis Type: In FM-2D v5.2606, six analysis types are available (see Figure 4.14):

- **Eigenvalue Analysis:** Eigenvalue (modal) analysis to obtain the fundamental periods of the building and the corresponding mode shapes.
- **Static Pushover:** Static linear/nonlinear pushover analysis using a pre-specified mode pattern.
- **IDA:** Incremental dynamic analysis (IDA) up to collapse (Vamvatsikos and Cornell 2002).
- **Dynamic: Target scale factor:** Linear/Nonlinear response-history analysis by scaling a set of ground motion records using a pre-specified constant scale factor.
- **Dynamic: Target intensity:** Linear/Nonlinear response-history analysis by scaling a set of ground motion records to a pre-specified target seismic intensity at a pre-specified period.
- **ELF:** Linear/Nonlinear static analysis using a pre-specified equivalent lateral force profile.



Figure 4.14. Analysis parameters module: Analysis options tab

4.4.2 Run Options tab

The run option tab includes the following options (see Figure 4.15):

Show animation: select this option to display the structural deformation profile during analysis. If selected, the user has the option to specify the deformation profile scale factor as well as the size of the animation window in pixels. It is not recommended to use this option when running IDA for large buildings, as it can increase the run time.

Show openSees status: select this option to display *OpenSees* feedback during analysis. This will be displayed within the Windows Command Shell that will open when FM-2D is running. This option can be very useful in tracking possible problems with the *tcl* code.

Show scope: select this option to display a live plot of the maximum SDR and PFA recorded during the analysis.

Maximum runtime: specify the maximum run time for a pushover analysis or a single ground motion record analysis. This option is useful for terminating infinitely iterative loops that may occur during simulations due to convergence problems. A value of 5 to 10 minutes is recommended in most cases.

Include modeling uncertainty: in the case of response-history analysis, select this option if you need to include the numerical modeling uncertainty in your analysis. In that case, you need to specify the logarithmic standard deviation associated with the different numerical modeling parameters such as those of the employed deterioration component models or the material and damping parameters, as shown in Figure 4.16. Note that the current version 5.2606 assumes that all parameters are log-normally distributed and uncorrelated. As part of the uncertainty consideration, a Monte Carlo scheme is employed where many numerical model realizations are generated and analyzed for each ground motion record.

Run using parallel cores: this option is available for dynamic analysis procedure (not including IDA). It is used to run several dynamic analysis simulations (under multiple records or several realizations) in parallel using the available processing cores. This option saves significant amount of time depending on the number of utilized processing cores, as specified by the user. Note that this option is still in the beta version and parallel execution may not run on some computers.

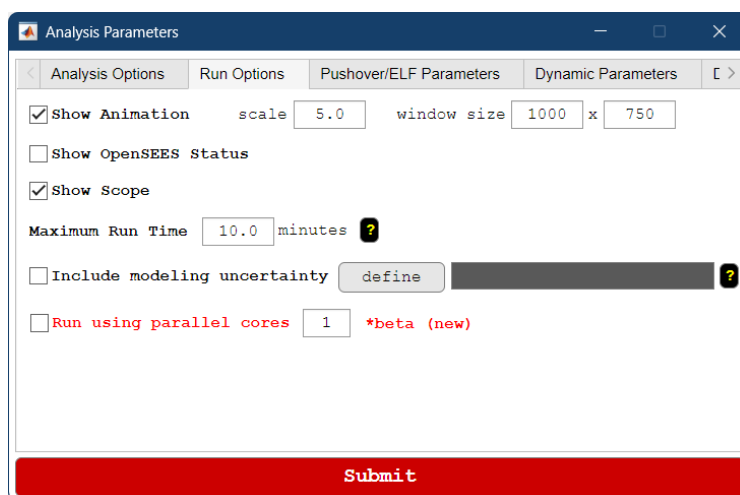


Figure 4.15. Analysis parameters module: Run options tab

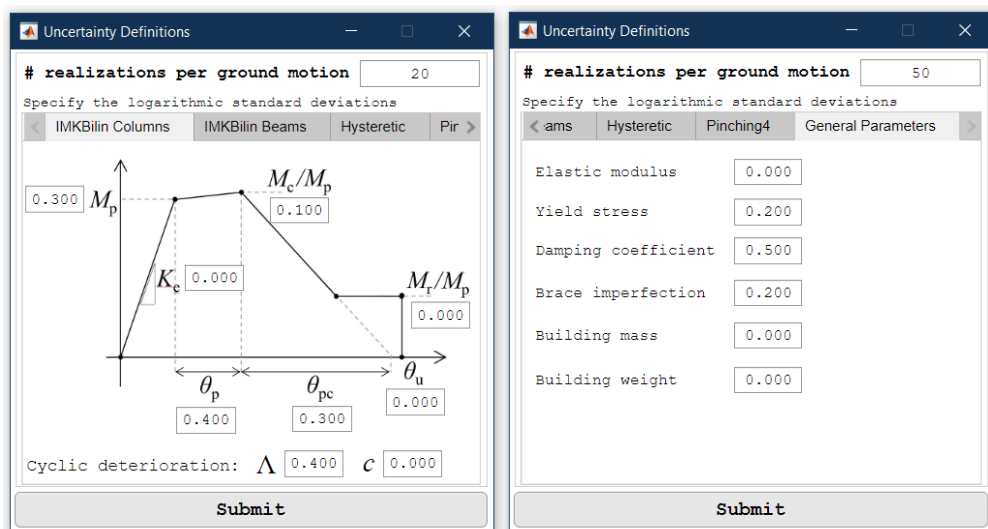
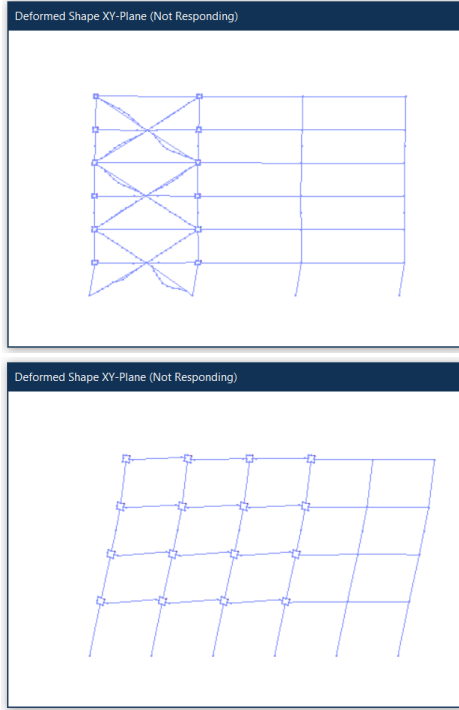
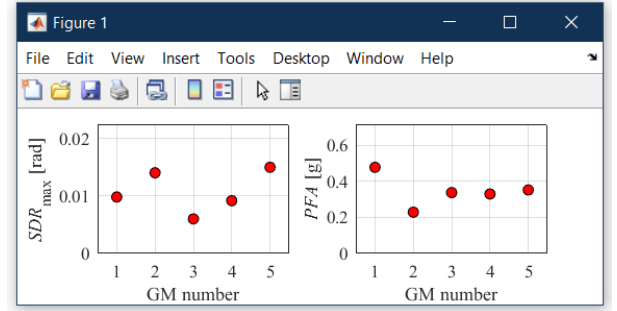


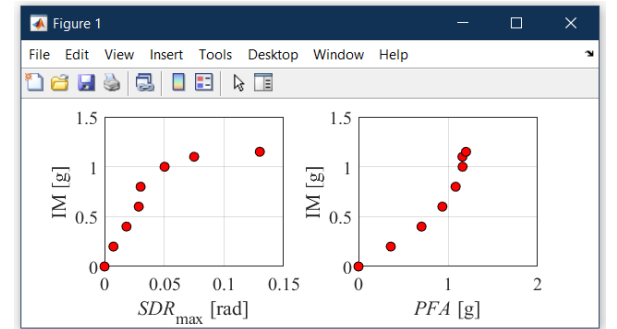
Figure 4.16. Module for modeling uncertainty definition



(a)



(b)



(c)

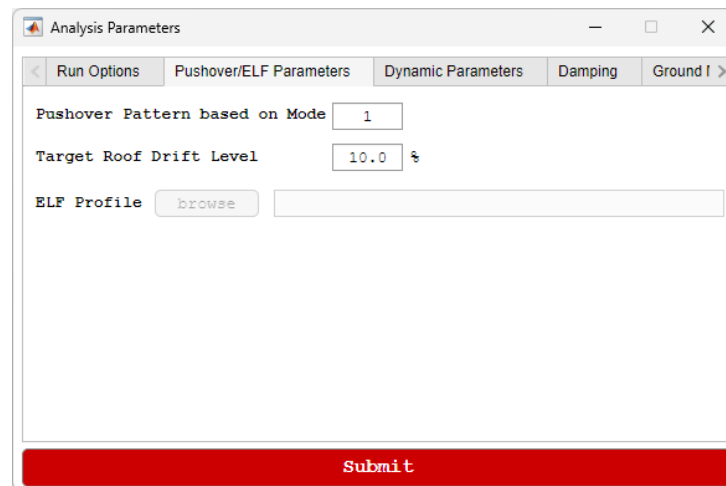
Figure 4.17. (a) Deformed shape window; (b) scope figure for dynamic analysis; (c) scope figure for IDA

4.4.3 Pushover/ELF Parameters tab

Pushover pattern based on mode: specify the Eigenmode number for which the modal shape will be used as a displacement pattern for pushover analysis.

Target roof drift level: specify the target drift of the roof (top floor) as a percentage of the total building height.

ELF profile: browse for a **.txt* file containing a single column of data representing the values of the equivalent lateral forces to be applied at each floor. Note that for a building with n number of stories, the n number of forces needs to be specified in the text file representing all building floors except for the ground floor.



The screenshot shows a software window titled "Analysis Parameters" with a tabbed interface. The "Pushover/ELF Parameters" tab is selected. It contains three input fields: "Pushover Pattern based on Mode" with a value of 1, "Target Roof Drift Level" with a value of 10.0 and a percentage symbol, and "ELF Profile" with a "browse" button and an empty text box. A large red "Submit" button is at the bottom.

Figure 4.18. Analysis parameters module: Pushover/ELF parameters tab

4.4.4 Dynamic Parameters tab

Intensity measure: for any type of dynamic analysis, select the intensity measure type to be used in ground motion scaling. Two types of acceleration-based intensity measures are available in the current release: 1) the spectral acceleration at the building first-node period, $Sa(T_1)$, and 2) the spectral acceleration ordinates average (geometric mean) within a given range of periods Sa_{avg} (Eads et al. 2015).

Scaling period: specify the reference period that will be used to scale ground motion records.

Scaling factor: specify the constant scaling factor that will be used to scale ground motion records.

Target intensity: specify the target seismic intensity, in units of g, for which the ground motion records' intensity measure will be scaled.

Adaptive intensity increment: for IDA, select this option to let FM-2D change the seismic intensity increment (step) depending on how close the analysis is to the collapse point. This option can help save run time for IDA analysis compared to using a constant increment.

Initial increment: for IDA, specify the initial intensity increment if an adaptive intensity increment is selected. If not, then specify the constant intensity increment.

Collapse point tolerance: for IDA, specify the tolerance (accuracy) to be used when tracking the collapse point; that is the error for the identified collapse intensity will be within +/- the tolerance.

Collapse drift limit: for dynamic analysis, specify additional collapse-identification criteria based on the story-drift ratio limit as to which collapse can be assumed to have occurred. In general, deteriorating systems will collapse under increasing P-Delta forces before reaching this SDR limit.

Maximum IDA curve slope: for dynamic analysis, specify additional collapse-identification criteria based on the maximum slope of the SDR-versus-Intensity IDA curve, beyond which collapse can be assumed to have occurred. A value of 0.4rad/g is recommended (corresponding to an increase of 0.04 radians per an intensity increment of 0.1g).

Analysis Parameters

< Pushover/ELF Parameters Dynamic Parameters Damping Ground Motions Reco >

Intensity Measure $Sa(T_1)$, $\xi = 5.0\%$

Scaling Period 1.218 [sec]

Scaling Factor 2.000

Target Intensity 0.990 [g]

☒ Adaptive Intensity Increment Initial Increment 0.20 [g]

Collapse Point Tolerance 0.05 [g]

Collapse Drift Limit 0.150 [rad]

Maximum IDA Curve Slope 0.500 [rad/g]

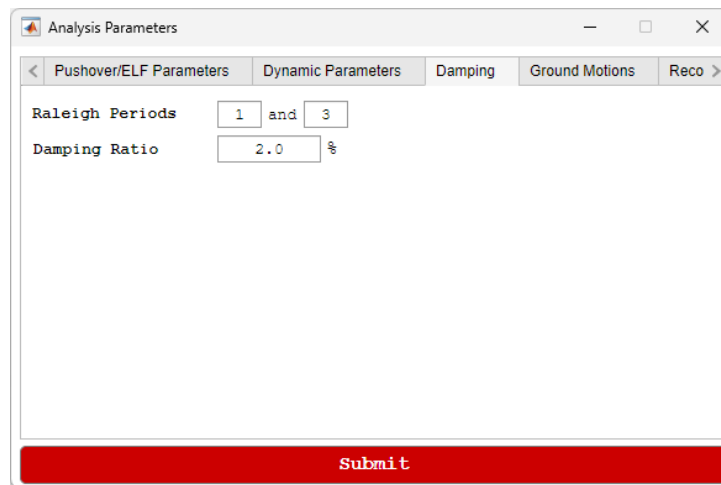
Submit

Figure 4.19. Analysis parameters module: Dynamic parameters tab

4.4.5 Damping tab

Raleigh Periods: for dynamic analysis, specify the periods used in the Raleigh damping model.

Damping Coefficient: for dynamic analysis, specify the damping coefficient as a percentage.



The screenshot shows a software window titled "Analysis Parameters" with a standard Windows-style title bar (minimize, maximize, close buttons). Inside the window, there is a tabbed interface with five tabs: "Pushover/ELF Parameters", "Dynamic Parameters", "Damping", "Ground Motions", and "Reco". The "Damping" tab is currently selected and highlighted. Within this tab, there are two input fields. The first is labeled "Raleigh Periods" and contains two text boxes with the values "1" and "3", separated by the word "and". The second is labeled "Damping Ratio" and contains a text box with the value "2.0" followed by a percentage symbol "%". At the bottom of the window, there is a prominent red button labeled "Submit".

Figure 4.20. Analysis parameters module: Damping tab

4.4.6 Ground Motions tab

Ground Motion Folder: for dynamic analysis, browse the folder containing the ground motion records. This folder should include individual ASCII text files for ground motion records. The text file should include the acceleration data in units of g. The data can be organized in a single or multiple-column format. The folder should also include a file named *AllGMInfo.txt*. This file should include two columns of data: the record name (also the acceleration file name) and the time step. Note that once the ground motion folder is selected, the ground motion data will be processed and stored internally in the project's MATLAB file. In that sense, the GM folder need not be present when the project file is transferred to another machine.

Consider GM Records: for dynamic analysis, specify the range of records to be considered. This option is meant to allow the user to run a subset of the imported records.

Free Vibration Time: for dynamic analysis, specify the amount of free vibration time to be added at the end of each ground motion record. A value of 10 seconds is typically used to allow the structure to come to rest and to trace residual drifts.

Analysis Time Step: for dynamic analysis, specify the size of the time step for the transient analysis as a function of the ground motion record time step. A value of 0.25 to 0.5 is commonly used.

Inspect GM: this button will open an interactive module where the user can explore the time histories of the imported GM records and investigate the records various properties such as the frequency content and effective duration.

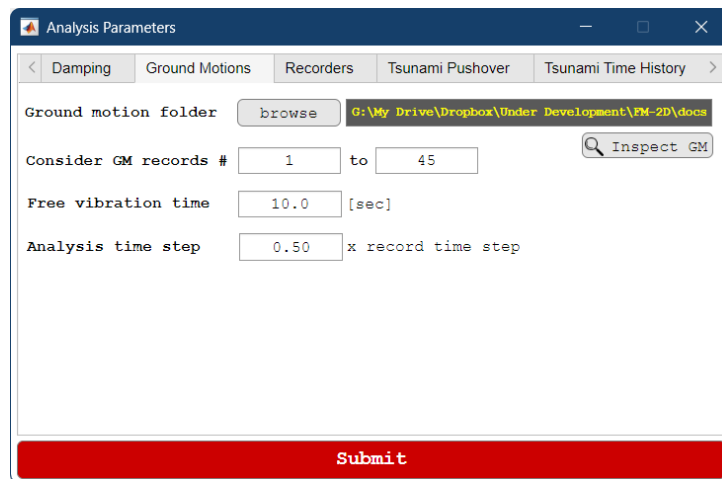


Figure 4.21. Analysis parameters module: Ground motions tab

4.4.7 Recorders tab

From this tab, the user can select the local and global EDPs to be saved. By default, the SDR, PFA and Eigen modes are saved. Using the checkboxes shown in Figure 4.22, other recorders can be activated. The description of each EDP and the name notation of the generated output files are summarized in Table 4.1.

Table 4.1. Recorder options and their definitions

Recorder Option	Description	Output Filename *
Time	Analysis time vector	Time.out
Floor Displacement	Floor absolute displacement	Disp[F]_MF.out
Story Drift Ratio	Story drift ratio	SDR[S]_MF.out
Floor Acceleration	Relative floor acceleration	RFA[F]_MF.out
Floor Velocity	Relative floor velocity	RFV[F]_MF.out
Column Element Forces	Axial, shear, and moment forces for elastic column elements	Column[S][A].out
Beam Element Forces	Axial, shear, and moment forces for elastic beam elements	Beam[F][B].out
Column Spring M/R	Moment and rotation of the column springs	ColSpring[S][A][L]_F.out ColSpring[S][A][L]_D.out
Beam Spring M/R	Moment and rotation of the beam springs	BeamSpring[F][B][L]_F.out BeamSpring[F][B][L]_D.out
Panel Zone Spring M/R	Moment and rotation of the panel zone springs	PZ[F][A]_M.out PZ[F][A]_R.out
Brace F/D	Axial force and deformation of the brace elements	Brace[S][B]_F.out Brace[S][B]_D.out
Floor Link F/D	Axial force and deformation of the floor links truss elements	FloorLink[F]_F.out FloorLink[F]_D.out
Support Reaction	Reaction forces at ground supports	Support[A].out
Eigen Modes	Eigen mode periods	EigenPeriod.out
Eigen Vectors	Eigen mode vectors	EigenVectorsMode[M].out

* [F] Floor number, [S] Story number, [B] Bay number, [A] Axis number, [M] Mode number

[L] Spring location: For Beams: Top (T) or Bottom (B) spring within a story
For columns: Left (L) or Right (R) spring within a bay

Analysis Parameters

< Damping Ground Motions **Recorders** Tsunami Pushover Tsunami Time History >

☒ Time	☒ Column Spring M/R	☒ Eigen Modes
☒ Floor Displacement	☒ Beam Spring M/R	☒ Eigen Vectors
☒ Story Drift Ratio	☒ Panel Zone Spring M/R	
☒ Floor Acceleration	☐ EGF Connection Spring M/R	
☐ Floor Velocity	☐ Floor Link F/D	
☒ Column Element Forces	☐ Brace F/D	
☐ Beam Element Forces	☐ Corner Gusset Plate Force	
☒ Support Reaction		
☐ Delete output data files after finishing the run		

Submit

Figure 4.22. Analysis parameters module: Recorders tab

5. Visualization Options

The visualization module, shown in Figure 5.1, provides the user with a variety of options to visualize the analysis results. The visualization options differ based on the frame type, the analysis procedure, and the recorded EDPs. The plot options are described below.

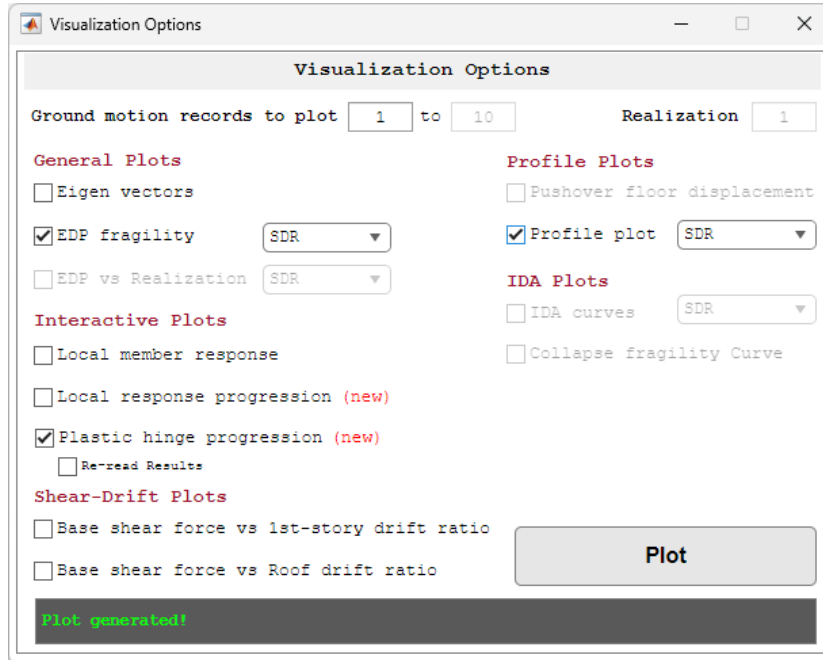


Figure 5.1. Visualizations options

Eigen Vectors: Plots the profile of the Eigen vector for the different mode periods as shown in Figure 5.2.

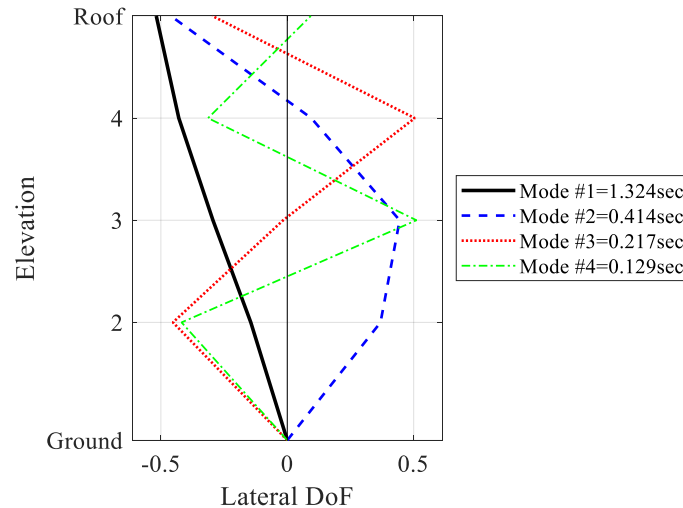


Figure 5.2. Eigenvectors plot

Shear-Drift Response: Plots the base shear force, normalized by the seismic weight, versus the first-story drift ratio or the roof drift ratio. Figure 5.3a and Figure 5.3b show sample plots of the base shear-drift response based on dynamic and pushover analysis, respectively. Note that in the case of dynamic analysis, the user can specify, from the top of the visualization module, the range of ground motion records to be processed and plotted.

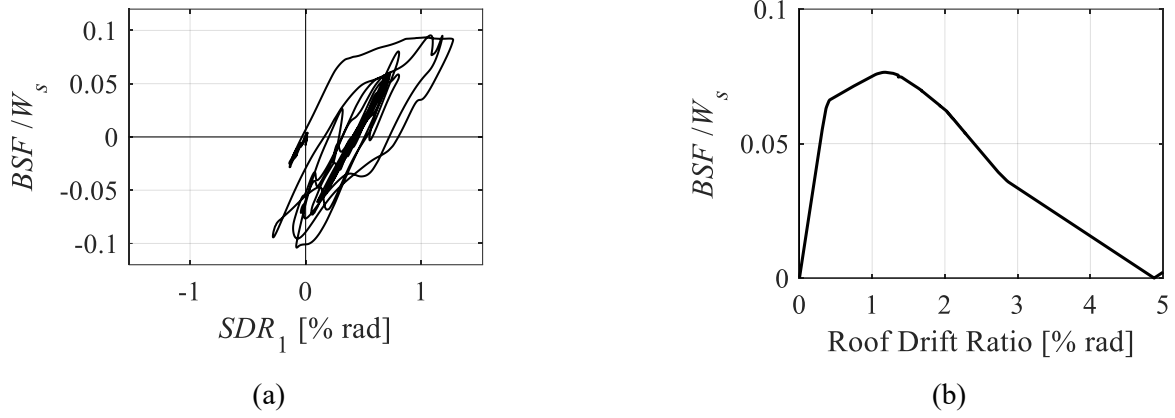


Figure 5.3. Normalized base shear force plots: (a) versus first-story drift ratio based on dynamic analysis; (b) versus roof drift ratio based on pushover analysis

IDA Plots: In the case of IDA, two additional plots can be generated. The first plot is the IDA curves; which is a plot of the seismic intensity versus a specific EDP as shown in Figure 5.4a for the SDR . IDA curves are generated for SDR , RDR , PFA , and PFV unless the recorder was not activated for any of those EDPs. The second plot is the collapse fragility curve as shown in Figure 5.4b. This curve is a lognormal cumulative distribution function (CDF) that is fitted to the empirical probability distribution of the identified collapse intensities for the different ground motion records.

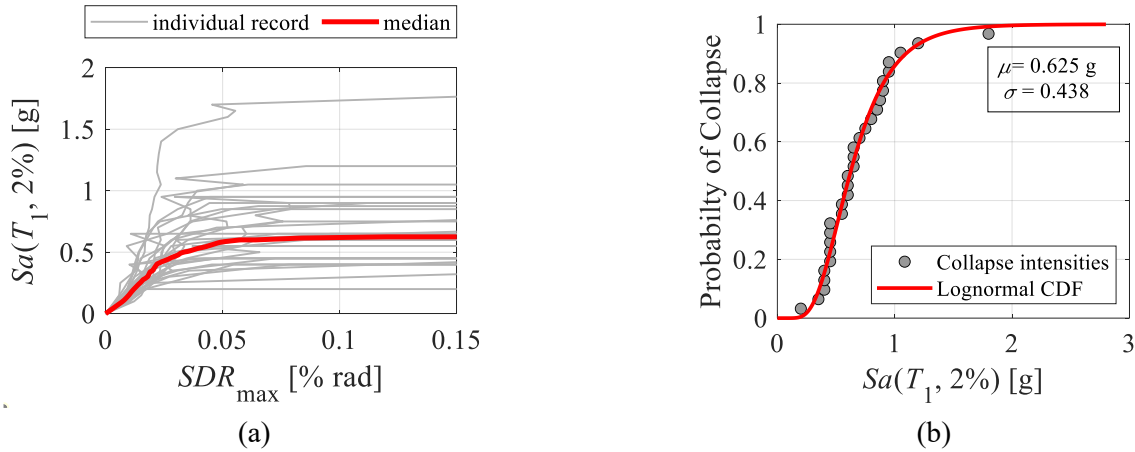
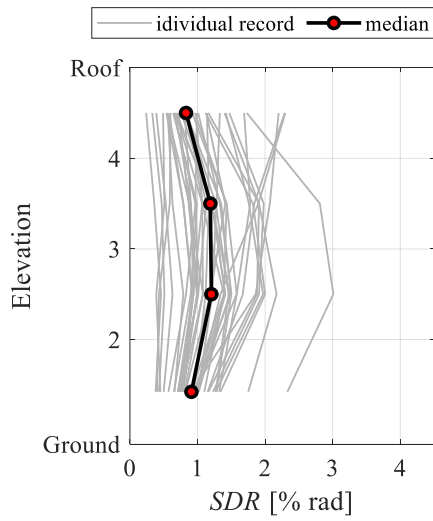
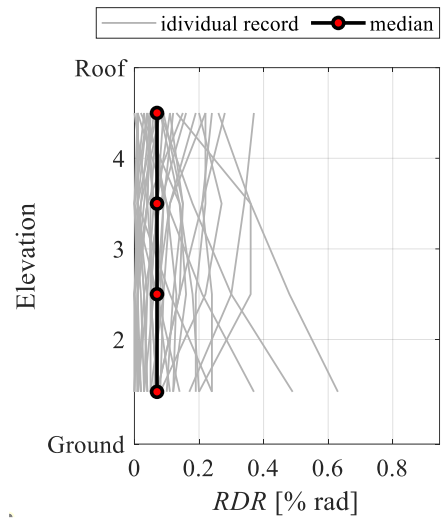


Figure 5.4. IDA plots: (a) IDA curve for maximum SDR ; (b) collapse fragility curve

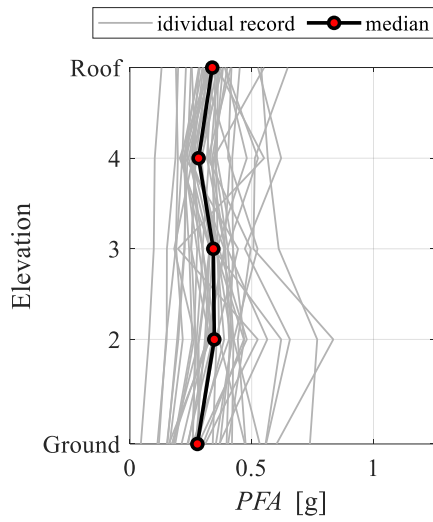
EDP Profiles: Plots the profile of the different EDPs along the building height. Figure 5.5 shows the different profile plots generated by FM-2D for SDR , RDR , PFA , PFV , and Displacement profile. As shown in these figures, in the case of multiple ground motion records (more than 3), the median profile is computed and superimposed.



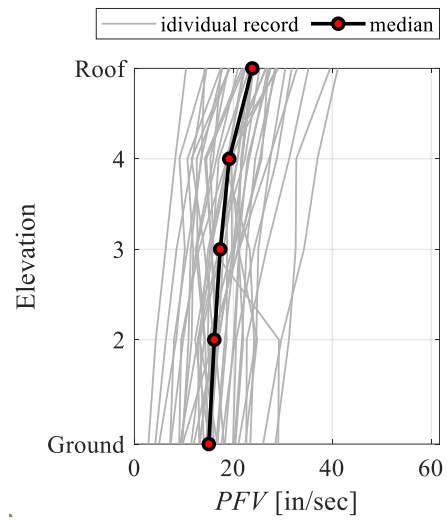
(a) Story-drift ratio



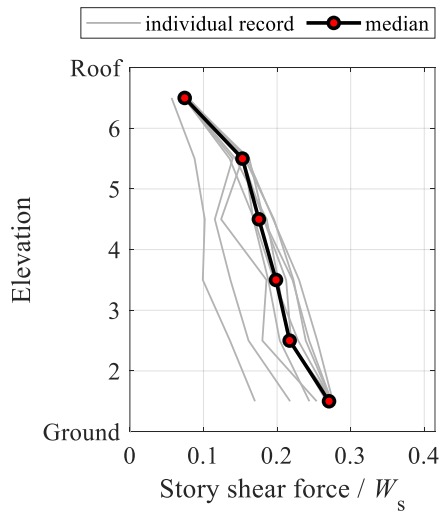
(b) Residual story-drift ratio



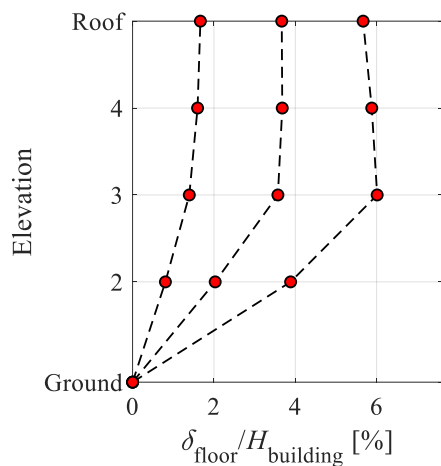
(c) Peak floor accelerations



(d) Peak floor velocity



(e) Story shear force



(f) Displacement profile for Pushover/ELF analysis

Figure 5.5. EDP profile plots

Local member response: This will open an interactive interface as shown in Figure 5.6. From this interface, the user can explore the responses of the frame elastic elements and nonlinear springs that were selected to be recorded.

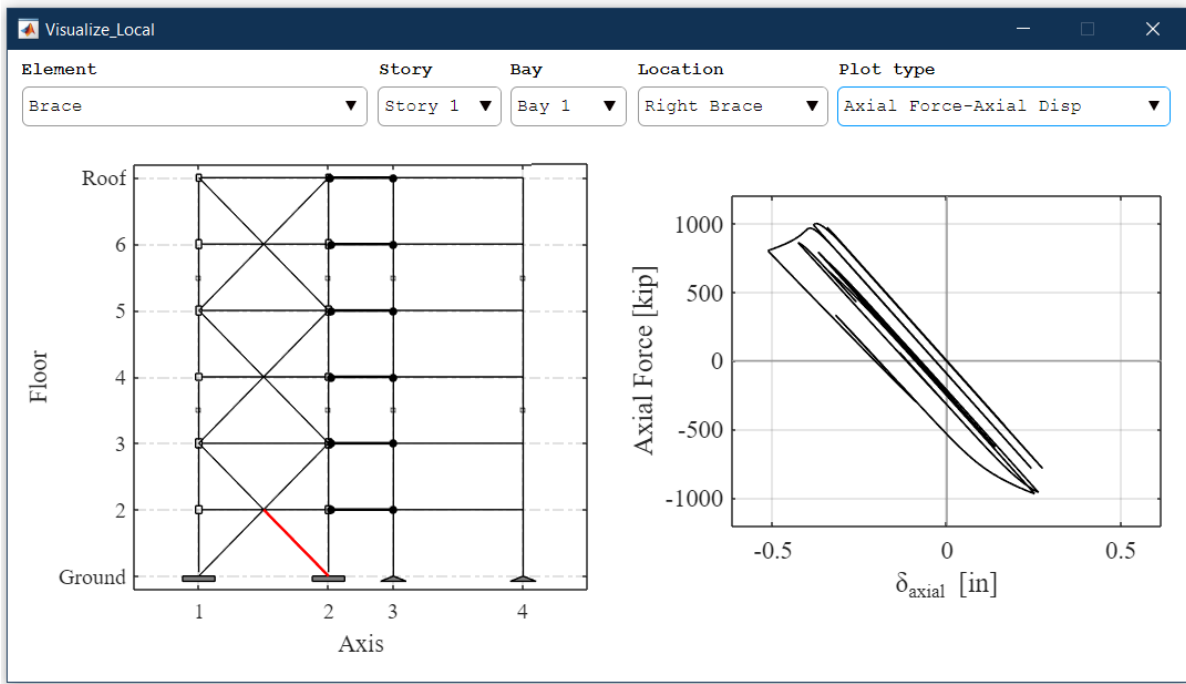


Figure 5.6. Interactive interface for visualizing local member response

Local response progression: This will open an interactive interface as shown in Figure 5.7. From this interface, the user can explore the progression of a local member or a spring response in relation to the global (Normalized base shear versus 1st story drift) response. A slider tool is provided in this interface to scroll through the steps of the analysis while visualizing the step location on the global response. Using checkboxes, the user can choose to snap quickly to the point of maximum base shear or the point of maximum drift. Note that this interface is valid for all analysis types.

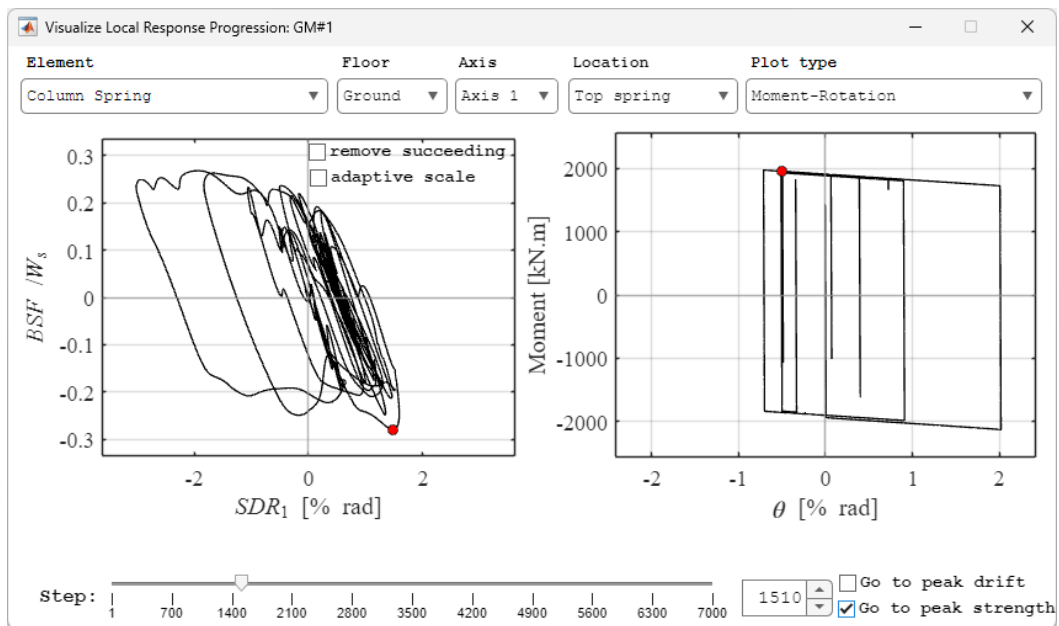


Figure 5.7. Interactive interface for visualizing the local response progression

Force diagrams: This will open an interactive interface as shown in Figure 5.8. From this interface, the user can explore the distribution of axial, shear, and moment forces in the beams and columns throughout the analysis. The user can control the size of the diagrams and the font.

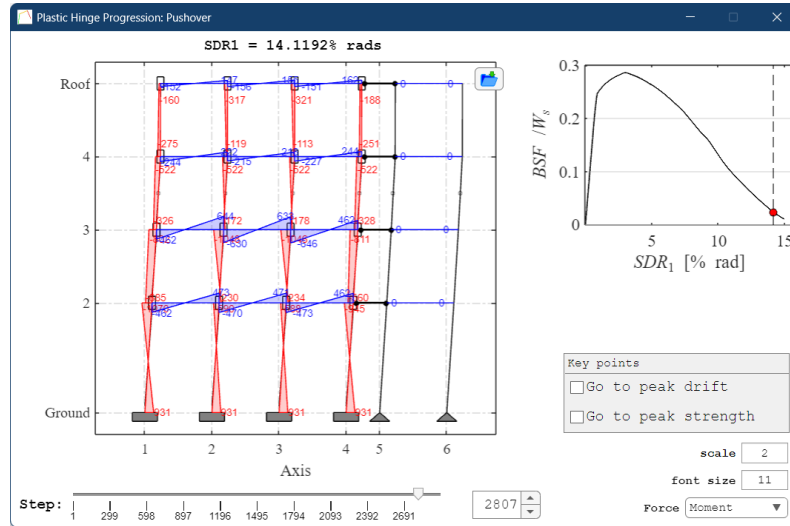


Figure 5.8. Interactive interface for visualizing the frame force diagrams

Plastic hinge progression: This will open an interactive interface as shown in Figure 5.9. From this interface, the user can explore the progression of plastic hinge development in all the rotational springs throughout the analysis. Three color-coded states are used to define the status of any rotational spring: 1) green marker means the spring is in the elastic stage, 2) orange marker means the spring is in the plastic (post-yield) stage, and 3) red marker means the spring is in the post-peak (strength degradation) stage. A slider tool is provided in this interface to scroll through the steps of the analysis while visualizing the step location on the global response (Normalized base shear versus 1st story drift). Using checkboxes, the user can choose to snap quickly to the point of maximum base shear or the point of maximum drift. Note that this interface is valid for all analysis types.

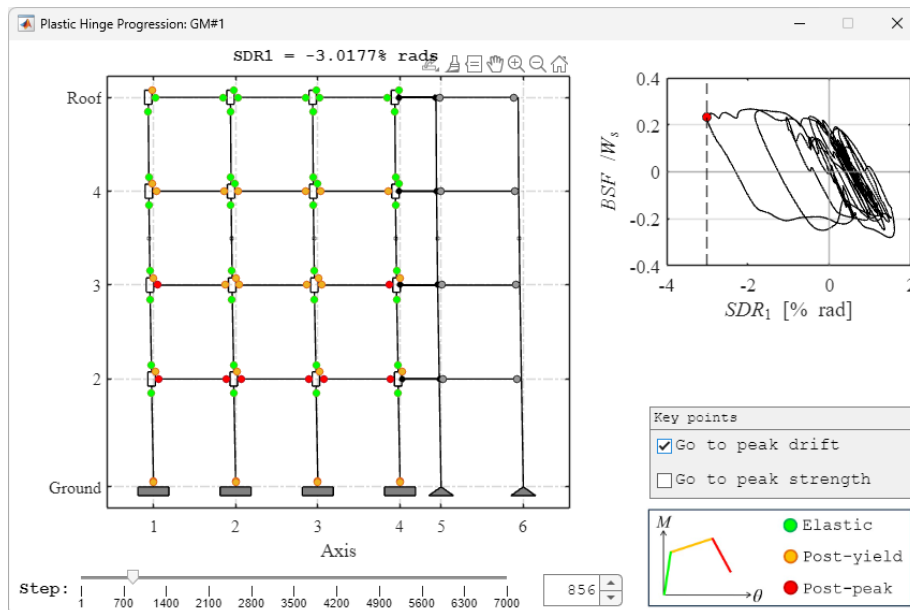


Figure 5.9. Interactive interface for visualizing plastic hinge progression

6. Project Inspector

The user can verify the defined project components and parameters by using the “scope” button located in the project file management panel (see Figure 4.1). This button will open a window that provides a summary of the current project definition as shown in Figure 6.1.

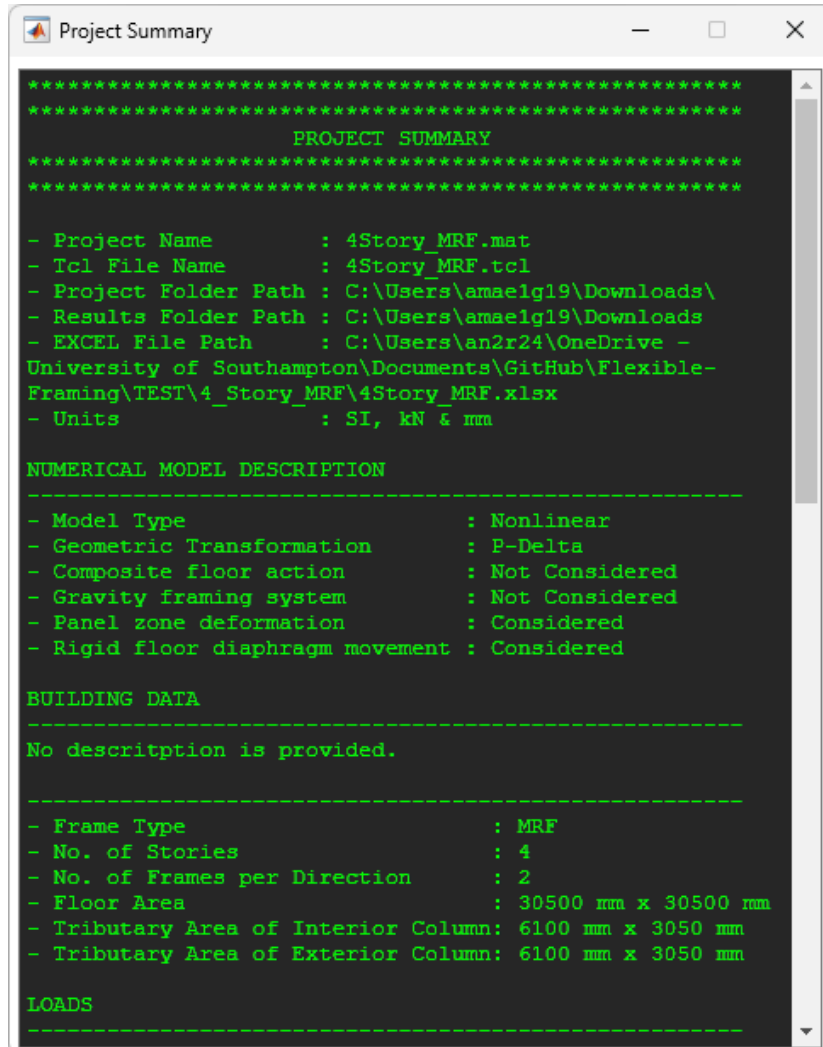


Figure 6.1. Project summary window

7. Modeling Methodology

FM-2D employs state-of-the-art numerical modeling recommendations when constructing the numerical models. A brief description of the main modeling assumptions is provided below.

7.1 Main Frame Steel Columns

The main frame columns are modeled as elastic elements with nonlinear rotational springs at their end (i.e., lumped plasticity approach). The springs are assigned the modified Ibarra-Medina-Krawinkler (IMK) phenomenological deterioration model (Ibarra et al. 2005; Lignos and Krawinkler 2011). In particular, the recently updated [IMKBilin](#) model. The moment and rotation parameters of this model's characteristic monotonic backbone curve (see Figure 7.1a) as well as the cyclic deterioration parameters, are defined based on the latest recommendations by Lignos et al. (2019) for wide-flange columns and by Hartloper et al. (2026) for HSS columns.

7.2 Steel Moment Beam-to-Column Connections

Similar to the columns, the flexibility of the main frame beam-to-column moment connections are modeled as nonlinear rotational springs. The springs are assigned the modified Ibarra-Medina-Krawinkler (IMK) phenomenological deterioration model (Ibarra et al. 2005; Lignos and Krawinkler 2011). In particular, the recently updated [IMKBilin](#) model is used. The moment and rotation parameters of this model's characteristic monotonic backbone curve (see Figure 7.1a) as well as the cyclic deterioration parameters, are defined based on the recommendations by Lignos and Krawinkler (2011).

7.3 RC Beams and Columns

RC beams and columns are idealized as linear elastic beam-column elements with zero-length nonlinear rotational springs at their ends. The elastic elements can be modeled as either fiber-based elements or elastic beam-columns. The rotational springs are assigned the modified Ibarra-Medina-Krawinkler (IMK) deterioration model with a peak-oriented hysteretic response (Ibarra et al. 2005). The parameters of the backbone curve of this model are computed as per the empirical regression equations developed by Haselton et al. (2008). Note that these models do not account for undesirable shear failure that is expected in older non-ductile RC buildings.

7.4 Steel Column Panel Zone

If the column panel zone deformation is considered (see Section 4.2.1: Modeling Options), is explicitly modeled using the “parallelogram” model as discussed in Gupta and Krawinkler (1999) as shown in Figure 7.2a. A rotational spring allocated at the corner of the idealized panel zone is used to capture the moment-shear distortion response. The rotational spring is assigned the [Hysteretic](#) material model. Referring to Figure 7.1b, the three points defining this model backbone curve (that is the panel zone stiffness, strength and shear distortion parameters) are defined as per the latest recommendations by Skiadopoulos et al. (2020) that were developed as an update to the original Krawinkler (1978) model. If the composite action is considered (see Section 4.2.1: Modeling Options), the bare steel

strength parameters of the panel zone are modified as per Elkady and Lignos (2014). Note that this model is meant for wide-flange (or I-shaped) sections. For other section types, modifications are needed to the model.

7.5 Braces and Gusset Plates

The steel brace model is based on the force formulation proposed by Spacone et al. (1996). The brace component is divided into two elements that are offset (initial camber) in the mid-length of the brace to trigger global (flexural) buckling. A displacement-based formulation is used where the brace is discretized by n segments (8 segments by default). The displacement-based beam-column element is assigned a corotational transformation to account for large displacements due to global buckling. A fiber-based approach is employed to represent the brace cross-section. The cross-section fiber elements are assigned a uniaxial engineering stress-strain relationship that is based on the Menegotto-Pinto [Steel02](#) model (Menegotto and Pinto 1973). Depending on the cross-section type (rectangular or circular Hollow Structural Section (HSS) or wide-flange) a different discretization process and set of material parameters are employed based on the recommendations by Karamanci and Lignos (2014) based on extensive calibrations with experimental data. Fracture resulting from low-cycle fatigue is simulated by masking the *Steel02* material with the *Fatigue* material model (Uriz et al. 2008). This material model is based on a linear strain accumulation rule following the Coffin-Manson relationship (Manson 1965). In summary, this modeling approach explicitly simulates the post-buckling brace behavior and the interaction between the second order bending moment and the axial force, while capturing the spread of plasticity along the brace depending on the steel brace cross-section and the onset of ductile fracture.

The flexibility and flexural yielding of the gusset plates due to out-of-plane brace bending are explicitly simulated using a nonlinear out-of-plane rotational spring placed at the ends of each brace as proposed by Hsiao et al. (2013).

7.6 Gravity Framing

If the gravity framing system is considered (see Section 4.2.1: Modeling Options), the equivalent gravity frame is used to represent it as discussed in Gupta and Krawinkler (1999). At each story, the flexural strength and stiffness properties of the columns as part of the equivalent gravity frame are set equal to half of the summation of those of the interior gravity framing. Similarly, the equivalent gravity beams per floor are assigned flexural strength and stiffness equal to half of the summation of those of the gravity beams in the considered orthogonal direction.

The gravity beam-to-column connections are assumed as conventional shear-tab connections (a single steel plate fillet welded to the supporting column with a single column of structural bolts). A rotational spring is used to simulate the moment-rotation response of these connections. This spring is assigned the *Pinching4* deterioration model which simulates a deteriorating pinched load-deformation response. The model's backbone curve is defined based on four points as shown in Figure 7.1c. The four points (i.e., moment and rotation parameters) for bare steel and composite shear-tab connections are defined based on the recommendation by Elkady and Lignos (2015). The *Pinching4* model is masked with a [MinMax](#) model to superimpose the ultimate rotations at failure (assumed to be 8% and 12% radians for bare and composite connections, respectively). Furthermore, the user has the option to consider the sudden increase in the connection's stiffness and strength due to beam binding on the column flange when the gap between

the two closes (refer to Elkady and Lignos (2015)). This will involve adding another spring in parallel which is assigned the elastic perfectly-plastic gap model, [ElasticPPGap](#).

7.7 Damping

Damping is simulated with the Rayleigh model. The damping coefficient, ζ , is assigned to two modal periods; all of which are defined by the user (see Section 4.4.5: Damping tab). The Rayleigh damping stiffness-proportional term is assigned only to the elastic column and beam elements while the mass-proportional term is assigned to all the frame nodes with masses as suggested by Zareian and Medina (2010). The proposed method of assigning damping to the frame model helps avoiding fictitious damping forces when the frame behaves in the nonlinear range.

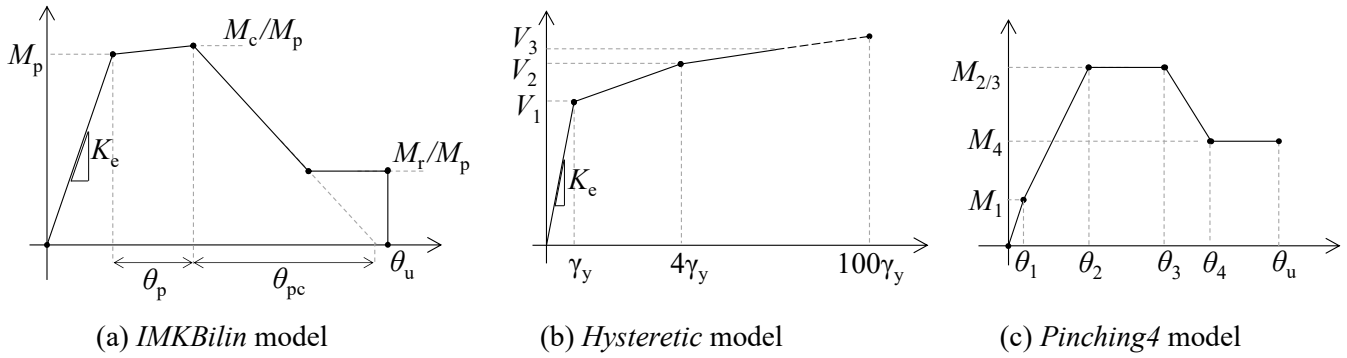


Figure 7.1. The backbone curve parameters for the main deterioration models used in FM-2D

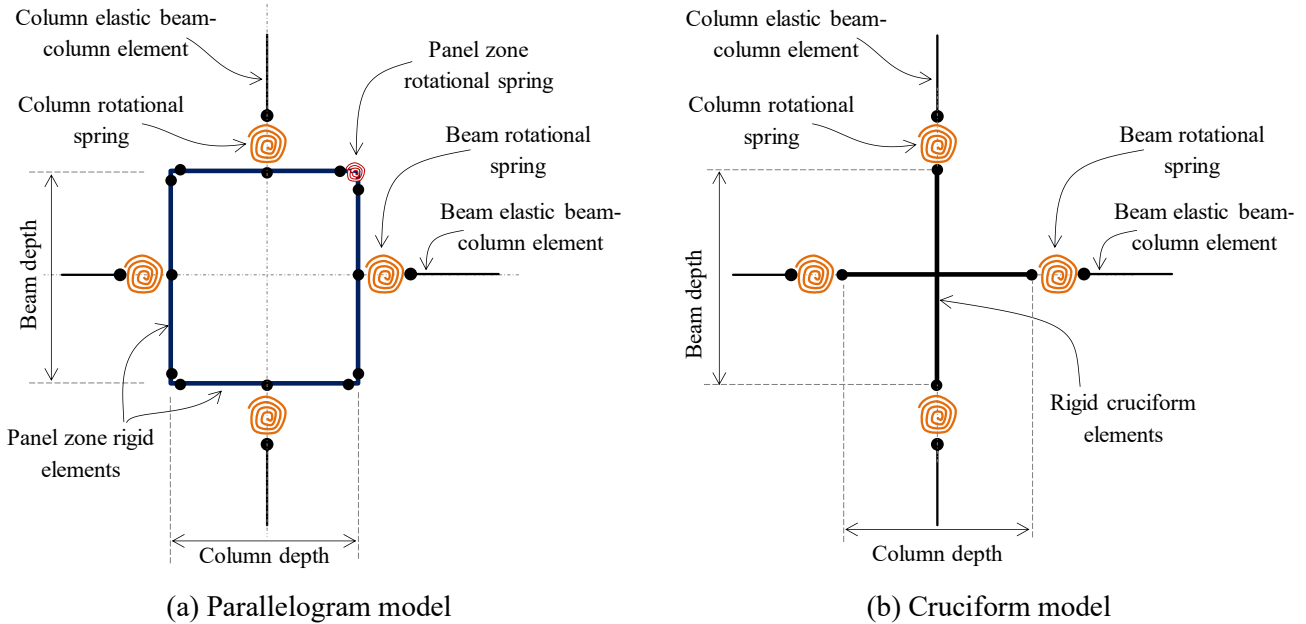


Figure 7.2. Types of panel zone idealization when (a) considered and (b) ignored

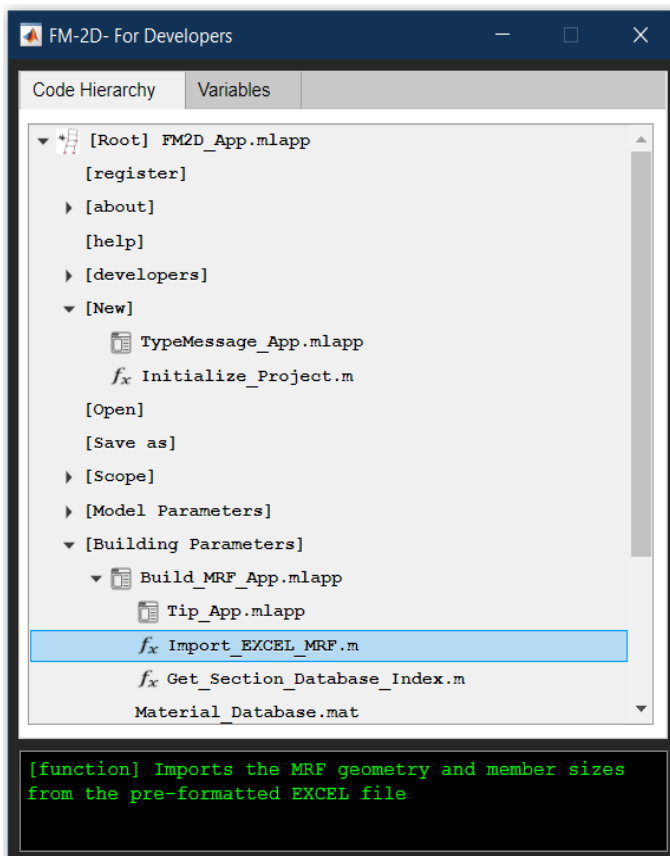
8. For Developers

FM-2D is fully developed in MATLAB. The entire source code is a combination of MATLAB's **.mlapp* graphical user interface (GUI) files, **.m* script files, and **.mat* data files as well as *OpenSees* **.tcl* subroutine files. The source code is available in the online GitHub repository inside the folder named *src*. Users who are interested in further developing FM-2D can do so through the GitHub platform. In brief, the user should 1) Create an account on GitHub.com, 2) Download and install the GitHub Desktop application, 3) Fork FM-2D's repository (i.e., take a parallel copy of FM-2D's code on the user's local computer) and 4) finally, further developing and expanding the source code. Proposed changes to the code should then be publicly shared with other users, under the GPLv3 license. The user can "commit" any changes to the online repository where they will be vetted and validated before being patched to the master code. Credit will be given to any user contributions in newly compiled executables. Consult the GitHub quick start guide [here](#).

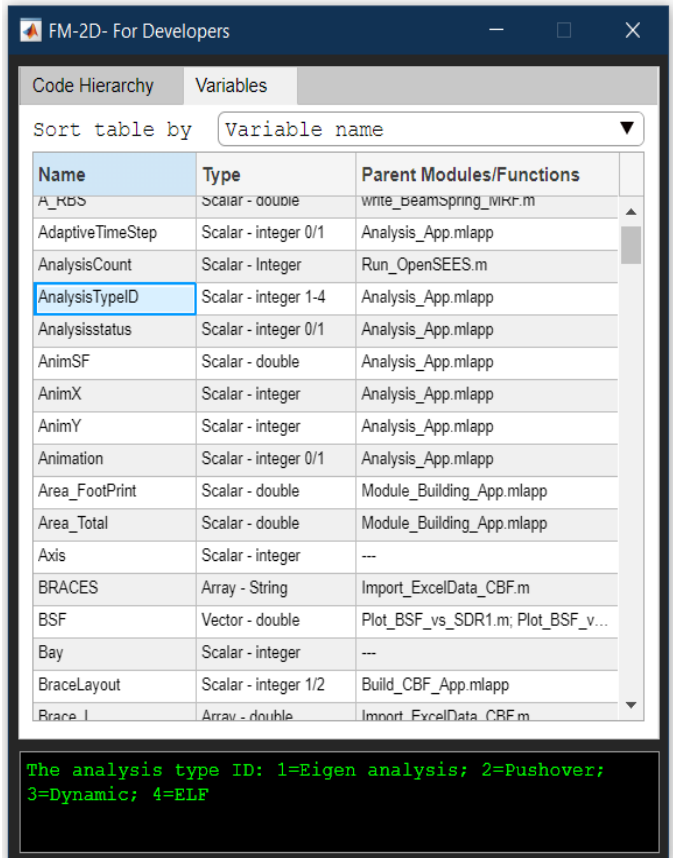
To help users understand the source code, the MATLAB script code within the **.m* function files is well-commented and organized. Furthermore, by clicking the "developers" button in FM-2D's main console, the user is prompted to the interface shown in Figure 8.1. . This interface comprises two tabs:

- (a) Code Hierarchy tab, see Figure 9.1a, shows the code's organizational tree. By expanding the branches of this tree list, the user can inspect the dependency and the relation between the different source code GUI and function files. Additionally, by clicking on a specific element (GUI or function name) within this code tree, a full description of this element is displayed in the tip window at the bottom of the interface. The description also includes the names of the output variables generated by a specific function.
- (b) Variables tab, see Figure 9.1b, which lists the details of all the variables used within the source code, including their name, type (scalar, vector, matrix, structure, etc.), and the parent module(s) or function(s) where this variable is used and defined. By clicking on a given variable name, a full description of the variable will be displayed in the tip window.

These tools are aimed at helping the user comprehend the source code. As such, the user is potentially able to navigate the source code and modify/add specific functions accordingly.



(a)



(b)

Figure 8.1. Developers interface for source code hierarchy and variable definitions

8.1 Project Variables

To improve project data organization and management, the many project variables are consolidated within few structured variables. Specifically, when loading the project *.mat file in Matlab, 13 structured variables are loaded. The description of each variable is summarized in Table 8.1. Figure 8.2 shows the variables consolidated within each structured variable. The full description of each variable can be found in the developers interface as discussed in the next section.

Table 8.1. Structured variable description

12) Variable	13) Initiation location	14) Description
15) PROJECT	16) FM2D_App.mlapp	17) Stores variables defined at project initiation such as the frame type, units, directory for results, OpenSees filename, etc.
18) MODEL	19) Model_App.mlapp	20) Stores variables that control the general numerical modeling assumptions and the beam/column modeling approach.
21) LAYOUT	22) Build_App.mlapp	23) Stores variables defined within the building module that describe the building layout and geometrical parameters.
24) CONNECTION	25) Build_App.mlapp	26) Stores variables defined within the building module that describe the connection types, supports, and boundary conditions.
27) LOAD	28) Build_App.mlapp	29) Stores variables defined within the building module that describe the load cases and combinations.
30) MATERIAL	31) Build_App.mlapp	32) Stores variables defined within the building module that describe the material properties.
33) PROPERTY	34) Build_App.mlapp	35) Stores the variables exported from the Excel sheet; i.e., the MF dimensions and sections sizes.
36) RECORDERS	37) Analysis_App.mlapp	38) Stores the flags for the data to be recorded as defined within the analysis module.
39) FILENAME	40) Analysis_App.mlapp	41) Stores the root names of the recorder output files.
42) SIGMA	43) Analysis_App.mlapp	44) Stores the uncertainty (standard deviation) values as defined within the analysis module.

45) ANALYSIS	46) Analysis_App.mlapp	47) Stores the primary variables defined within the analysis module; i.e., analysis type and its parameters.
48) DRAW	49) Build_App.mlapp	50) Stores the variables needed for the plot and visualization functions
51) STATUS	52) FM2D_App.mlapp	53) Stores the definition status flags for the different modules

PROJECT	LAYOUT	PROPERTY	LOAD	RECORDERS	SIGMA	ANALYSIS
☐ Version ☐ MainDirectory ☐ ProjectPath ☐ ProjectName ☐ OpenSEESFileName ☐ Units ☐ FrameType ☐ BuildingDescription ☐ resource_root ☐ RFpath ☐ ExcelFilePath ☐ ExcelFileName ☐ GMFolderPath ☐ GMFolderName	☐ BraceLayout ☐ DIM1 ☐ DIM2 ☐ nMF ☐ NStory ☐ NBay ☐ nGC ☐ nGB ☐ TAex1 ☐ TAex2 ☐ TAin1 ☐ TAin2 ☐ FootPrintArea ☐ Perimeter ☐ TA_MF ☐ TD_MF ☐ TribAreaEx ☐ TribAreaIn	☐ HStory ☐ WBay ☐ MF_COLUMNS ☐ MF_BEAMS ☐ GF_COLUMNS ☐ GF_BEAMS ☐ Doubler ☐ Splice ☐ Fs_Profile ☐ SecData_BU ☐ WBuilding ☐ HBuilding ☐ BRACES ☐ CGP_tp ☐ MGP_tp ☐ CGP_L123 ☐ MGP_L123 ☐ MGP_W ☐ MGP_RigidOffset ☐ CGP_RigidOffset ☐ Brace_L ☐ MGP_Lc ☐ CGP_Lc ☐ MF_SL ☐ CONNECTIONS ☐ MF_EEPCs ☐ SecData_Columnns ☐ SecData_Beams	☐ TypicalDL ☐ RoofDL ☐ TypicalLL ☐ RoofLL ☐ Cladding ☐ TypicalGL ☐ RoofGL ☐ cDL_W ☐ cLL_W ☐ cCL_W ☐ cGL_W ☐ cDL_M ☐ cLL_M ☐ cCL_M ☐ cGL_M ☐ LoadApplyOption	☒ Time ☒ Disp ☒ SDR ☒ Column ☒ ColSpring ☒ Beam ☒ BeamSpring ☒ PZ ☒ RFA ☒ FloorLink ☒ EGFconnection ☒ Brace ☒ CGP ☒ Support ☒ RFV ☒ EigenModes ☒ EigenVectors ☒ Delete_Flag	☐ IMKcol ☐ IMKbeam ☐ Pinching4 ☐ PZ ☐ fy ☐ fyB ☐ fyG ☐ zeta ☐ GI	☐ ExecutionOption ☐ g ☒ Uncertainty ☐ nRealizations ☐ EV ☐ PO ☐ ELF ☐ Dynamic_TargetSF ☐ Dynamic_TargetSA ☐ EQ ☐ CDPO ☐ TTH ☒ Animation ☐ MaxRunTime ☒ ShowOpenSeesStatus ☐ ModePO ☐ DriftPO ☐ DampModel ☐ DampModelJ ☐ zeta ☐ zetaSA ☐ GM ☐ TFreeVibration ☐ dtstep ☐ CollapseSDR ☐ dinundation ☐ Drag_Coeff ☐ ro_water ☐ CDPOpattern ☒ Parallel ☐ nCores ☐ Type ☐ AnimSF ☐ AnimX ☐ AnimY ☐ IDA ☐ SA_metric ☐ T1 ☐ coeff_T_lower ☐ coeff_T_upper ☐ SA_Step ☐ SAstepmin ☐ SF ☐ TargetSA ☐ IDAslopeLimit ☒ AdaptiveTimeStep ☒ ShowScope ☐ GM_Start ☐ GM_Last ☐ nGM
STATUS	CONNECTION	MATERIAL	FILENAME	DRAW		
☐ Model ☐ Build ☐ Analysis ☐ Run ☐ Sigma ☒ cLoad	☐ MFconnection ☐ MidSpanConstraint ☐ a ☐ b ☐ c ☐ LoadApplyOption ☐ SupportGFS ☐ Kshearstatus ☐ Support ☐ SpliceConnection ☐ Orientation ☐ EGFconnection ☐ k_beambacing ☐ FloorLink ☐ Fs ☐ Ks ☐ Comp_I ☐ Comp_I_GC ☐ trib ☐ ts ☐ coeff_cracked ☒ bond_slip	☐ Ec ☐ Er ☐ fc ☐ fyR ☐ fyBrace ☐ fyGP ☐ muC ☐ muR ☐ SteelMatID ☐ SteelMatType ☐ E ☐ fy ☐ fu ☐ mu0 ☐ fyP ☐ fuP ☐ fyb ☐ fub	☐ Disp ☐ SDR ☐ Column ☐ ColSpring ☐ Beam ☐ BeamSpring ☐ PZ ☐ RFA ☐ RFV ☐ FloorLink ☐ Brace ☐ CGP ☐ Support ☐ EGFconnection	☐ GridX ☐ GridY ☐ AXIS ☐ Elevation ☐ HalfElevation ☐ XTickLabel ☐ YTickLabel ☐ Column_Index ☐ Beam_Index ☐ Brace_Index ☐ XSecID ☐ Column_Data ☐ Beam_Data ☐ Brace_Data		

Figure 8.2. Variables stored within each structured variable

8.2 Source Code File Description

The source code for FM-2D is available inside the *src* folder in the GitHub repository. The source code comprises several files that can be categorized into four types as shown in Figure 8.3. Each of these is described below.

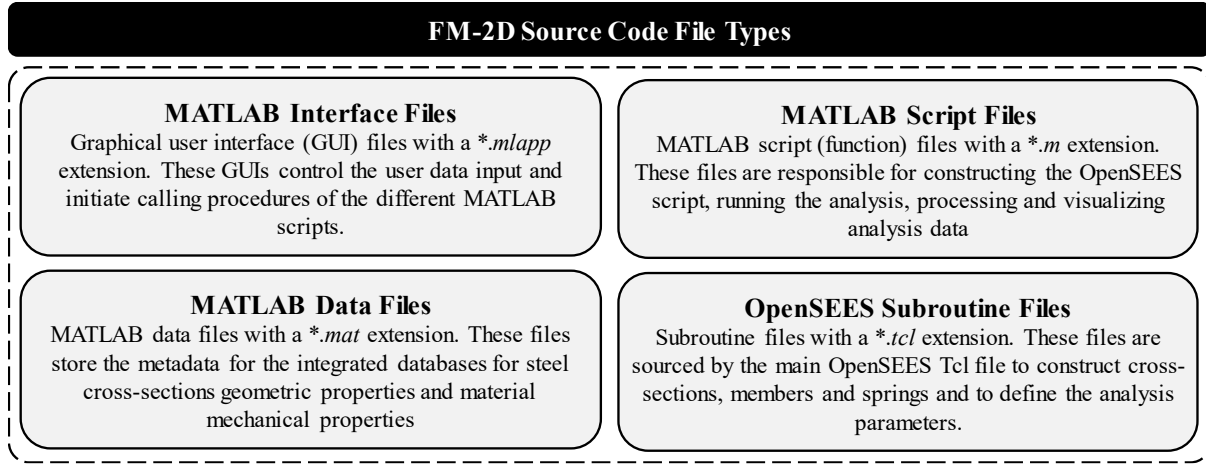


Figure 8.3. Types of source code files

8.2.1 MATLAB interface files

Those are the graphical user interface (GUI) files with a **.mlapp* file extension. These GUIs control the user data input and the subsequent calling procedures of the different subroutines and MATLAB codes. The files are:

FM2D_App.mlapp: Main console interface (see Figure 4.1).

TypeMessage_App.mlapp: Interface for selecting the frame type, the project's units, and the project's optional project description (see Figure 4.2).

Model_App.mlapp: Interface for defining the numerical model's parameters such as the geometric transformation and the consideration of composite action and gravity framing system contributions.

Build_App.mlapp: Interface for defining the building geometry, member sizes, loads, and materials.

Analysis_App.mlapp: Interface for defining the analysis type and its associated parameters.

ExecutionMessage_App.mlapp: Interface for defining the execution type; whether a new *tcl* file will be built for the run or if the run will be based on the existing *tcl* file.

Visulaize_App.mlapp: Interface for selecting the different visualization/plot options for the analysis results.

Visulaize_Local_App.mlapp: Interface for interactively visualizing the local response of the frame elements and springs.

Visulaize_Response_Progression_App.mlapp: Interface for interactively visualizing the local response progression in relation to global response.

Visualize_PH_Progression_App.mlapp: Interface for interactively visualizing the plastic hinge progression in the frame springs

Project_Summary_App.mlapp: Interface for displaying the project definitions summary when the “scope” button is clicked.

About_App.mlapp: Interface for displaying a brief description of FM-2D and the author's contact information.

Tip_App.mlapp: Interface for displaying a tip window when the help “?” button is clicked anywhere in FM-2D.

8.2.2 MATLAB script files

Those are the MATLAB code files with a *.m file extension. These files are used to build the *tcl* files, running the *OpenSees* analysis, processing and visualizing analysis data. This is a brief description of these script files:

Execution_Commands.m: Calls the different scripts for constructing the *tcl* files and then executes the *OpenSees* analysis

CREATOR_MODEL_MRF.m: executes the commands for generating the main *tcl* file for the steel MRF numerical model, excluding the analysis type code.

CREATOR_MODEL_CBF.m: executes the commands for generating the main *tcl* file for the CBF numerical model, excluding the analysis type code.

CREATOR_MODEL_MRF_RC.m: executes the commands for generating the main *tcl* file for the RC MRF numerical model, excluding the analysis type code.

CREATOR_ANALYSIS.m: generates the code lines in the main *tcl* file for the analysis procedures.

import_ExcelData.m: imports the MF geometry and member sizes from the pre-formatted EXCEL sheet.

import_GM_Data.m: imports the ground motion records acceleration and metadata from a specified folder.

write_*.m:** these scripts write specific lines of code to the main *tcl* file.

plot_*.m:** these scripts plot different aspects of the analysis results.

run_OpenSees.m: this is the main (root) script for running the different types of analysis, processing the analysis data, and saving summary results in a text file.

run_exe.m: runs the *OpenSees* executable.

read_Analysis_Data.m: read the dynamic analysis results from the *OpenSees* output files.

save_Increment_Data.m: save the incremental dynamic repose data in a structured variable *IncdData* such that it can be processed later.

save_Results_Dynamic.m: reads the results of dynamic analysis (SDR, RDR, and PFA) and saves a summary of the maximum absolute quantities in text files.

save_Results_IDA.m: reads the results of IDA (SDR, RDR, and PFA) that are stored in the *IncrData* variable and saves the IDA data in text files.

initialize_Project.m: Initialize some of the project parameters when a new project is created.

get_Mass_and_Period.m: gets the mass, and seismic weight and runs an Eigenvalue analysis to obtain the first mode period when the “Get period” button is clicked.

process_EigenAnalysis.m: processes the *OpenSees* Eigenvalue analysis output and stores a summary of the Eigenmodes (periods) and vectors in a text file.

load_SecData.m: loads the appropriate cross-section database, based on the member being modeled, and transforms the units of the metadata.

get_Section_Database_Index.m: finds and stores the index of the main frame member cross-sections within the database.

get_Project_Summary.m: collates the project summary text that will be shown when the “Scope” button is clicked.

get_Col_Axial_Force.m: computes the main frame columns’ compressive axial load demand based on the selected method; gravity load or pushover analysis load.

get_IM.m: computes the value of the intensity measures; $Sa(T_I)$ or Sa_{avg} .

get_SA_SDoF.m: computes the spectral acceleration of an elastic single degree of freedom system.

show_Status_Dynamic.m: shows, in the Windows Command Shell, a summary of the ground motion being analyzed, the seismic intensity, scaling factor, etc during dynamic analysis.

show_Scope_Dynamic.m: shows a live (instantly updated) plot of the SDR and PFA output during dynamic analysis.

show_Scope_IDA.m: shows a live (instantly updated) plot of the SDR and PFA IDA curves during IDA.

8.2.3 MATLAB data files

Those are the MATLAB data files with a *.*mat* file extension. These files store the metadata for the structural cross-section and materials properties. The files are:

AISC_WF_Sec_Database.mat: Contains a structured array with all the cross-section geometric properties based on the AISC wide-flange cross-section database. A total of 268 cross-sections are included ranging from W8 to W44. Note that data are stored in imperial units.

HSS_Sec_Database.mat: Contains a structured array with all the cross-section geometric properties based on the AISC hollow structural section cross-section database. A total of 519 circular and rectangular HSS cross-sections are included. Note that data are stored in imperial units.

EURO_I_Sec_Database.mat: Contains a structured array with all the cross-section geometric properties based on the European I-shaped cross-section database. A total of 504 cross-sections are included of type IPE, HEA, HEB, HL, HLZ, HD, UB, UBP, and UC. Note that data are stored in imperial units.

Material_Database.mat: Contains a structured array with all the mechanical properties of different steel types/grades as well as the corresponding calibrated parameters of the Voce-Chaboche constitutive plasticity model (von Mises 1913; Voce 1948; Chaboche 1989). The material parameters are based on extensive calibrations conducted by de Castro e Sousa and Lignos (2018); de Castro e Sousa et al. (2020). These parameters are used in FM-2D as part of the recently developed uniaxial UVC material model in *OpenSees* (Hartloper et al. 2019a; Hartloper et al. 2019b). Note that data are stored in imperial units.

8.2.4 OpenSees subroutine files

Those are the *OpenSees* subroutine (i.e., procedure) files with a *.*tcl* file extension. These files are called (i.e., sourced) by the main numerical model *tcl* file to construct different members and springs and to define the analysis parameters. The files are and their description are summarized in the table below.

Table 8.2. TCL subroutine files and their function description

TCL subroutine name	Subroutine function description
<i>Spring_Elastic</i>	Constructs a rotational spring with an elastic response
<i>Spring_Rigid</i>	Constructs a rotational spring with a large elastic stiffness (i.e., rigid).
<i>Spring_Zero</i>	Constructs a rotational spring with a zero stiffness (i.e., pin).
<i>Spring_Column_WideFlange</i>	Constructs a rotational spring with an <i>IMKBilin</i> response for wide-flange steel columns.
<i>Spring_Column_HSS</i>	Constructs a rotational spring with an <i>IMKBilin</i> response for HSS steel columns.
<i>Spring_Connection_FullyRigid</i>	Constructs a rotational spring with an <i>IMKBilin</i> response for beams as part of fully-rigid steel connections.
<i>Spring_IMK_RC</i>	Constructs a rotational spring with an <i>IMKPeakOriented</i> response representing the moment-rotation response of an RC beam-column.
<i>Spring_Pinching</i>	Constructs a rotational spring with an <i>IMKPinching</i> response.
<i>Spring_PZ</i>	Constructs a rotational spring for the column web panel zone with a trilinear hysteretic response.
<i>Spring_Gusset</i>	Constructs a rotational spring for the panel zone with a bilinear response based on Hsiao et al. (2013).
<i>FiberCHSS</i>	Constructs a fiber cross-section for a circular hollow structural steel section.
<i>FiberCHSS</i>	Constructs a fiber cross-section for a rectangular hollow structural steel section.
<i>FiberWF</i>	Constructs a fiber cross-section for a wide-flange/I-shaped cross-section.
<i>FiberRC_Rectangular</i>	Constructs a fiber cross-section for a rectangular RC cross-section.
<i>FiberRC_Circular</i>	Constructs a fiber cross-section for a circular RC cross-section.
<i>FatigueMat</i>	Constructs a <i>Steel02</i> material model masked with a <i>Fatigue</i> material to simulate ductile fracture.
<i>ConstructPanel_Rectangle</i>	Constructs the elements of the panel zone parallelogram model.
<i>ConstructPanel_Cross</i>	Constructs the elements of the panel zone cruciform model.
<i>ConstructFiberColumn</i>	Constructs a fiber displacement-based column element.
<i>ConstructBrace</i>	Constructs a fiber displacement-based brace element.
<i>Define_Material_RC</i>	Creates an RC uniaxial material using the <i>Concrete04</i> model.
<i>Generate_lognormrand</i>	Generates a log-normally distributed random variable for a specified mean and standard deviation.
<i>LibAnalysisStaticParameters</i>	Initiates the static analysis solver parameters.

<i>SolutionAlgorithm</i>	Defined the different convergence criteria parameters for static displacement-driven analysis.
<i>SolutionAlgorithmSub</i>	Iterates between the convergence criteria parameters defined in <i>SolutionAlgorithm.tcl</i> to find convergence in static displacement-driven analysis.
<i>SolverDynamic</i>	Runs dynamic (transient) analysis while alternating between solver algorithms and convergence test criteria to reach a solution.
<i>Check_SDRlimit</i>	Reads the story-drift ratio in each story at each time increment during the analysis and checks if the collapse SDR criteria were reached.
<i>DisplayModel3D</i>	Displays the node numbers, deformed or mode shape in all 3 planes.
<i>DisplayPlane</i>	Sets up the display parameters for specified view plane and display.

8.3 Model Node and Element Notation

Figure 8.4 to Figure 8.6 show the idealized MRF and CBF models along with the node and element notation that is being used in FM-2D. This notation needs to be respected by developers and users when modifying/adding to the *tcl* code, to avoid any potential overlap in material, element, section, and node IDs. In addition to these figures, the exact formulas used to generate the ID for the different *OpenSees* objects are defined below:

8.3.1 Nodes

Grid and support nodes:	$(10 \cdot F + A) \cdot 10$	
Beam end nodes:	$100 \cdot F + 10 \cdot A + i$	where, $i = [2,4]$
Column end nodes:	$100 \cdot S + 10 \cdot A + i$	where, $i = [1,3]$
Column splice nodes:	$100,000 + 1000 \cdot S + 100 \cdot A + i$	where, $i = [71,72]$
Beam plastic spring nodes:	$1000 \cdot F + 100 \cdot A + i$	where, $i = [20, 40]$
Panel zone nodes:	$400,000 + 1000 \cdot F + 100 \cdot A + i$	where, $i = [1,2,3,4,5,6,7,8,9,10,88,99]$
MGP rigid offset nodes:	$200,000 + 1000 \cdot F + 100 \cdot B + i$	
	where, $i = [1,2,3,4,5,6,7,12,13,14,15,16,17]$	for X-Bracing layout
	where, $i = [1,2,3,4,5,12,13,14,15]$	for Chevron layout
CGP rigid offset nodes:	$100,000 + 1000 \cdot F + 100 \cdot A + i$	
	where, $i = [40,41,50,51]$	for X-Bracing layout
	where, $i = [40,41]$	for Chevron layout
Brace segmental nodes:	$100,000 + 1000 \cdot F + 100 \cdot A + 41 + i$	where, $i = 1$ to n_{segment}
	$200,000 + 1000 \cdot F + 100 \cdot B + 16 + i$	where, $i = 1$ to n_{segment}
	$200,000 + 1000 \cdot F + 100 \cdot B + 17 + i$	where, $i = 1$ to n_{segment}

8.3.2 Elements

Column elastic elements:	$600,000 + 1000 \cdot S + 100 \cdot A + 41 + i$	where, $i = [0,1,2]$
Beam elastic elements:	$500,000 + 1000 \cdot F + 100 \cdot B + i$	where, $i = [0,1,2]$
Beam elastic offset elements:	$500,000 + 1000 \cdot F + 100 \cdot B + i$	where, $i = [2,4]$
Panel zone elements:	$7,000,000 + 1000 \cdot F + 100 \cdot A + i$	where, $i = [1,2,3,4,5,6,7,8]$
Floor link elements:	$1000 + F$	

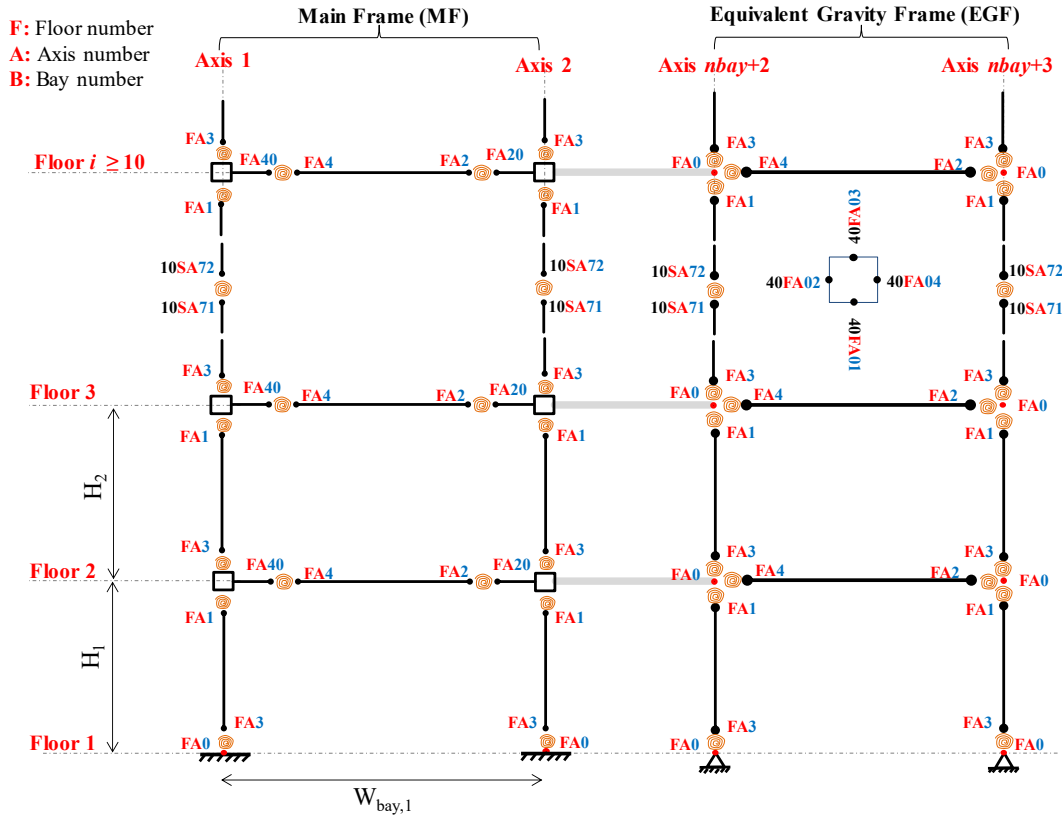
Brace elements:	$8,100,000 + 1000 \cdot S + 100 \cdot B$	
	$8,200,000 + 1000 \cdot S + 100 \cdot B$	
Brace ghost elements:	$4,100,000 + 1000 \cdot S + 100 \cdot B$	
	$4,200,000 + 1000 \cdot S + 100 \cdot B$	
MGP rigid elements:	$700,000 + 1000 \cdot S + 100 \cdot B + i$	where, $i = [22,33,44,55,66,77]$
CGP rigid elements:	$700,000 + 1000 \cdot S + 100 \cdot A + i$	where, $i = [11,99]$

8.3.3 Springs

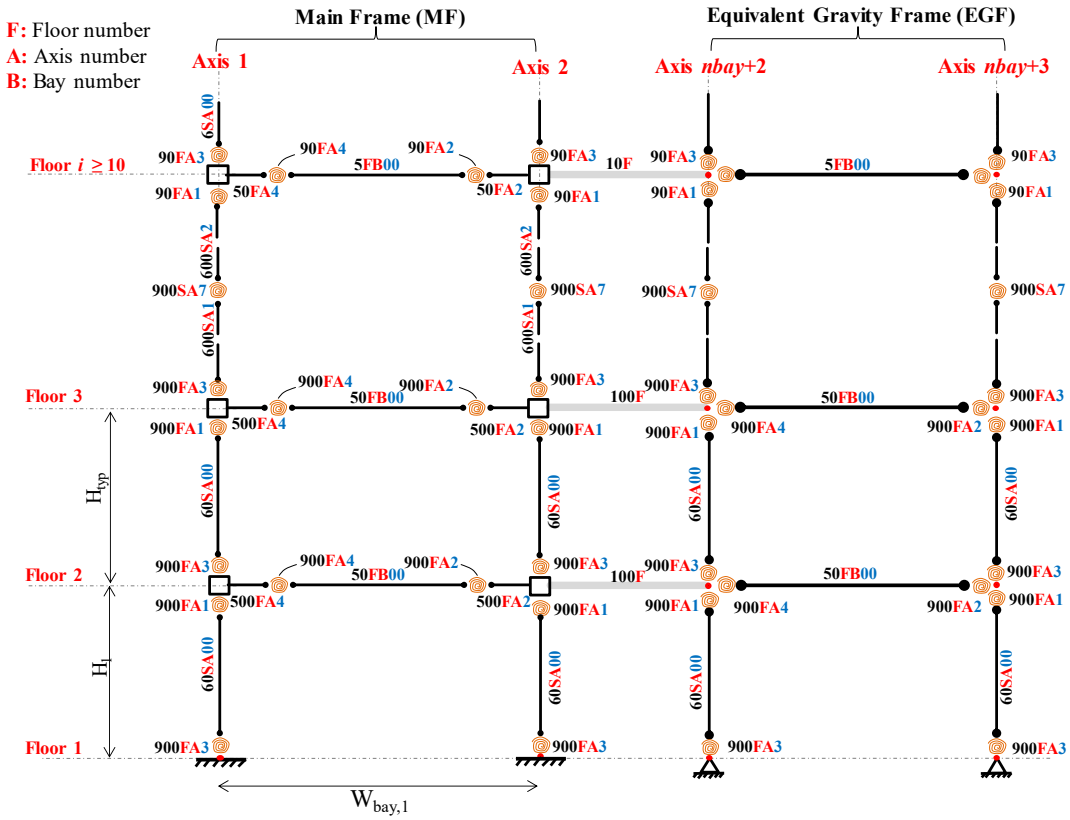
Column springs:	$900,000 + 1000 \cdot F + 100 \cdot A + i$	where, $i = [1,3]$
Beam springs:	$900,000 + 1000 \cdot F + 100 \cdot A + i$	where, $i = [2,4]$
Panel zone spring:	$900,000 + 1000 \cdot F + 100 \cdot A$	
Beam MGP springs:	$900,000 + 1000 \cdot F + 100 \cdot A + i$	where, $i = [22,55]$
Column splice spring:	$900,000 + 1000 \cdot S + 100 \cdot A + 7$	
MGP springs:	$900,000 + 1000 \cdot F + 100 \cdot B + i$	where, $i = [33,44,66,77]$
CGP springs:	$900,000 + 1000 \cdot F + 100 \cdot A + i$	where, $i = [11,99]$

8.3.4 Material

Flexible (weak) material:	9
Rigid material:	99
UVC material:	666
Fatigue material:	$100 \rightarrow 300$
HLB material:	$1000 \rightarrow 3000$

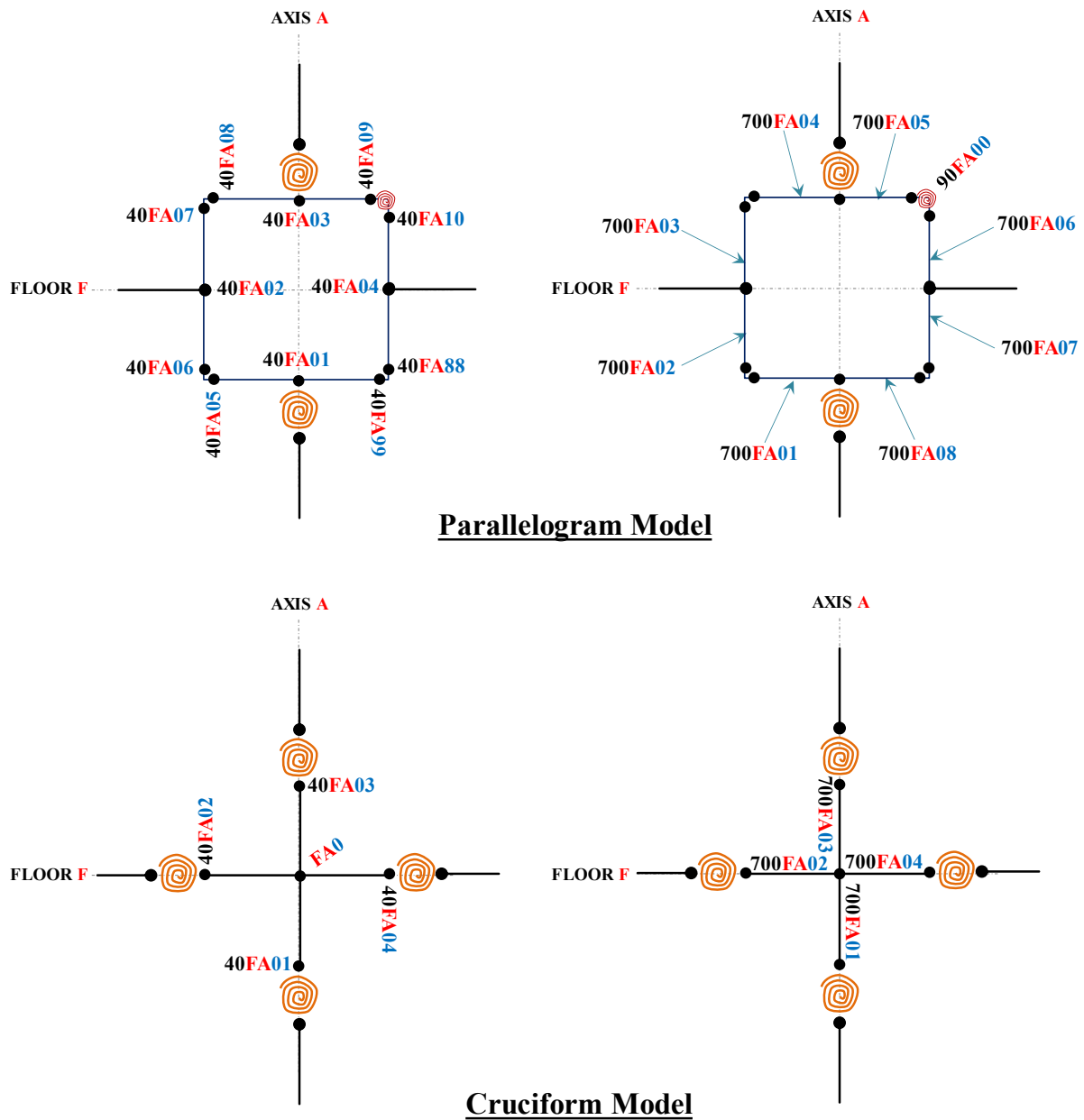


(a) Node notation



(b) Element notation

Figure 8.4. The node and element notation for the idealized moment-resisting frame model



(a) Node notation

(b) Element notation

Figure 8.5. The node and element notation for the idealized panel zone model

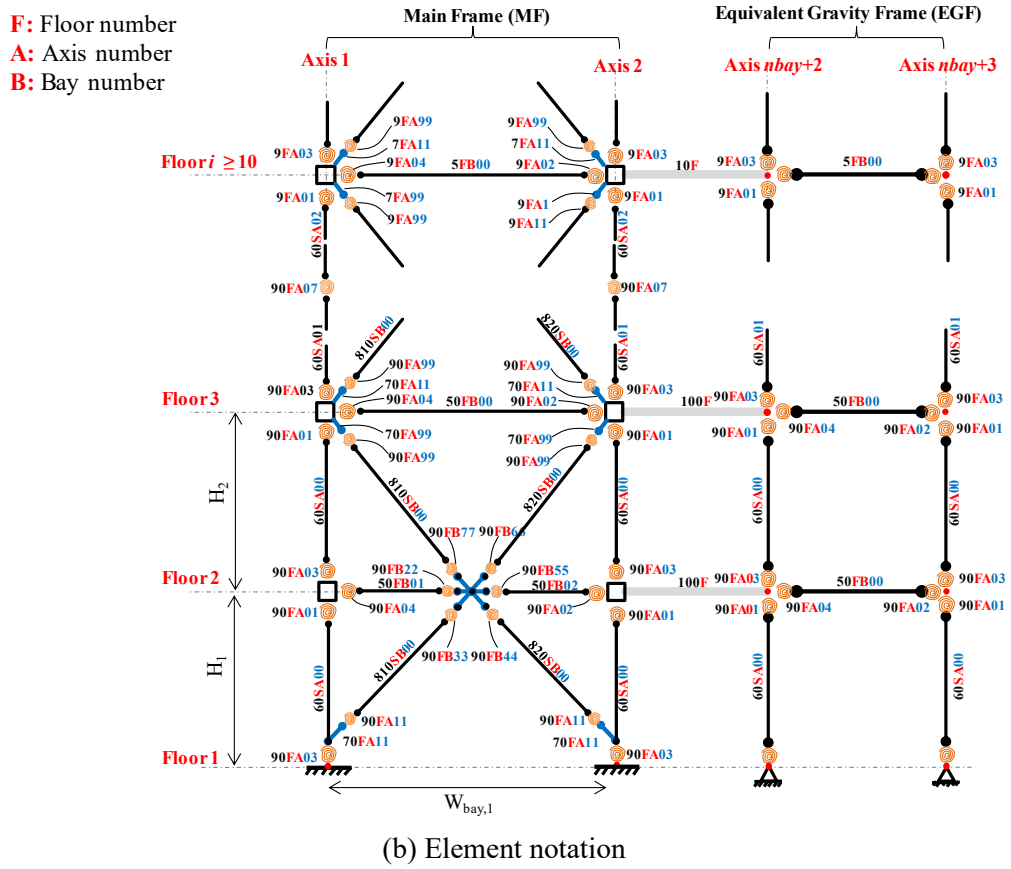
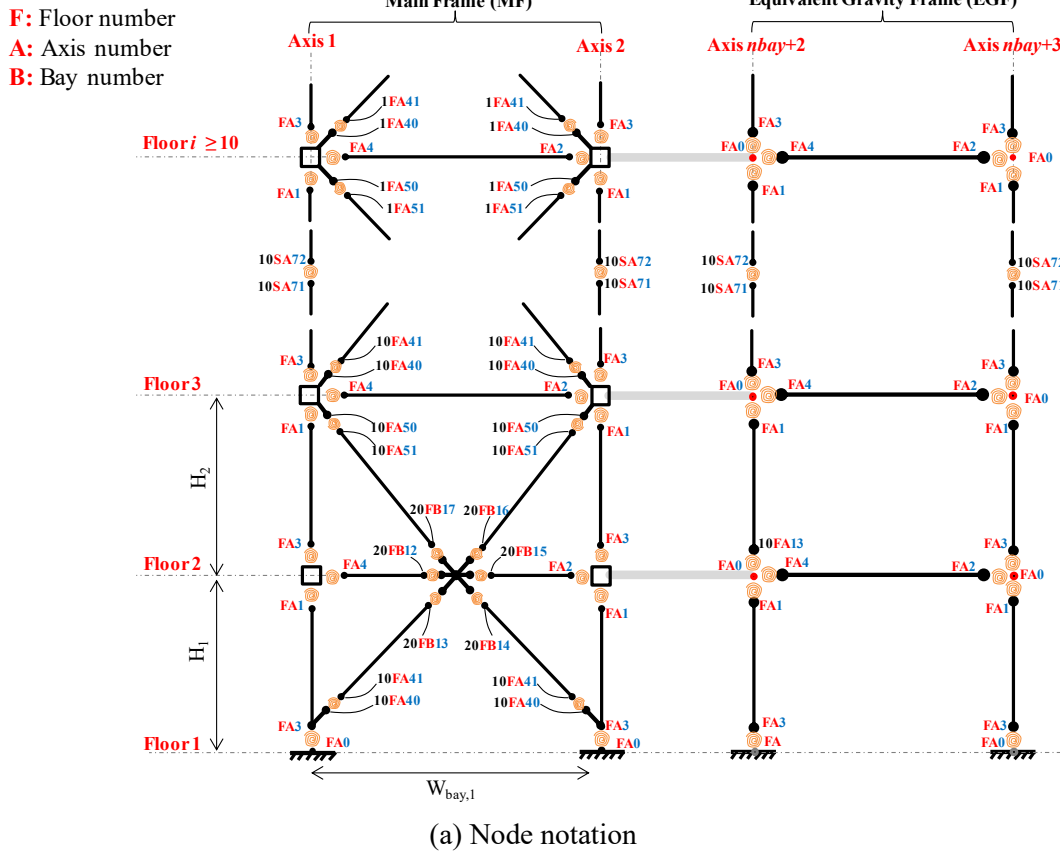


Figure 8.6. The node and element notation for the idealized concentrically-braced frame model

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